



ATHENS UNIVERSITY OF ECONOMIC AND BUSINESS

MSc in Business Analytics

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Modeling Expected Goals in Football

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Abstract

This study explores how to use statistical models to estimate the chance of scoring a goal in football, using the concept of Expected Goals (xG). Based on detailed data from the 2015–2016 Premier League season (provided by StatsBomb), the project examines key factors like shot distance, angle, player position, and defensive pressure. After data cleaning and Exploratory Data Analysis (EDA), four different logistic regression models were developed including a full model, two models after using AIC and BIC criteria, and a LASSO-regularized model to predict if a shot would result in a goal or not. Each model was evaluated using metrics like accuracy, precision, recall, F1-score, and ROC-AUC.

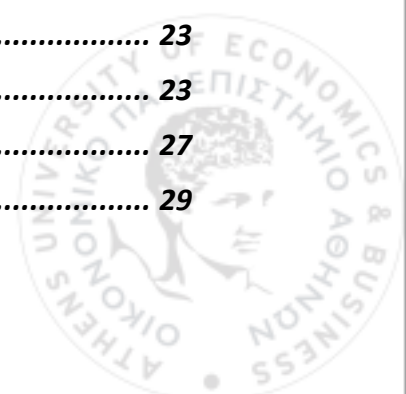
The results confirm that statistical modeling can successfully estimate scoring probability based on in-game variables. To validate the performance of the developed models, their predictions were compared to the official xG values provided by StatsBomb, highlighting both similarities and differences in estimation where we notice that the four models developed have strong predicting ability compared with this column.

These findings are useful for football coaches, performance analysts, and betting professionals who rely on data-driven decisions. Overall, the project highlights the growing role of data analytics in football and shows how xG modeling can support both strategy development and performance evaluation.



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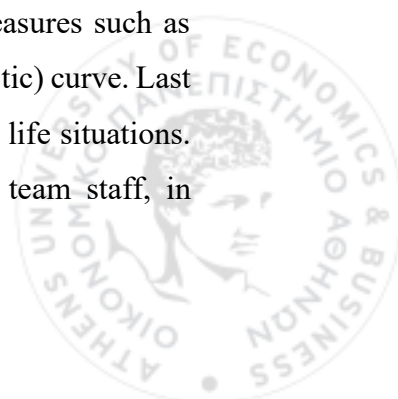


Chapter 1. Introduction

1.1 Purpose and Objectives

The goal of this paper is to identify and explain the challenges of using statistical models in football data analysis with a special focus on the Expected Goals (xG) prediction. Expected Goals (xG) is one of the most used metrics for assessing the quality of opportunities, and it provides a more unbiased picture of the performance of teams and players.

To begin with, let's define the concept of xG. It refers to the probability of a shot resulting in a goal, based on the conditions surrounding that moment. Understanding xG requires a theoretical foundation, as it depends on several influencing factors, such as the distance and angle of the shot, the positioning of defenders and the goalkeeper, the type of action (e.g., header or foot), and the nature of the pass leading to the shot. Defining these variables is essential to understanding how xG goes beyond simply counting shots, and instead attempts to evaluate the quality of each chance. It is therefore important to have a theoretical knowledge of this concept as it involves the analysis of the variables that affect it, including the range and direction of the ball, the structure of the defenders and the goalkeeper, type of action (header or foot), and the kind of pass that led to the last shot. It is therefore important to define these parameters in order to understand how the xG indicator tries to measure the 'quality' of a chance as opposed to just the number of shots or shots taken. The next step is to apply this theoretical understanding to real-world data. For this study, we used match data from the 2015/2016 Premier League season, extracted from the StatsBomb platform. Using this dataset, statistical models for xG prediction were built, combining logistic regression techniques after Exploratory Data Analysis (EDA). This approach helps to examine the individual contribution of each variable to scoring outcomes and to evaluate the predictive power of the model. Another purpose of this work is to assess the results of the applied statistical models. This is done by using measures such as accuracy, recall, precision and the ROC (Receiver Operating Characteristic) curve. Last but not the least, much emphasis has been put on the use of xG in real life situations. xG predictions can provide useful resources for coaches and other team staff, in

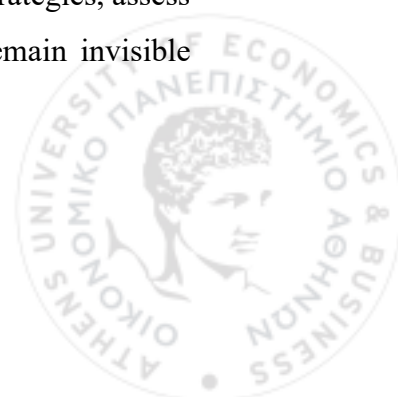


planning strategies and assessing the performance of individual players. At the same time, they also gain particular importance in the field of betting, in which betting companies use statistical models for setting odds and forecasting outcomes. The use of xG can result in better and more equitable predictions, which may influence the corporate policy of companies and the strategy of bettors. In this way, the work aims to illustrate the nature of xG analysis and explore the problems that come with the use of statistics in football.

1.2 The Importance of Data Analytics in Football

In the evolving landscape of modern football, data analysis has emerged as a powerful tool for gaining a competitive edge. From optimizing training sessions and evaluating player performance to enhancing match-day tactics and scouting talent, data-driven insights are transforming how teams operate. Technologies such as GPS trackers, heart rate monitors, and advanced tracking systems enable coaches and analysts to collect extensive information on player movement, effort, and decision-making, which in turn supports personalized training regimens, injury prevention strategies, and tactical refinement. Beyond the pitch, data analysis also influences recruitment and fan engagement. Clubs increasingly rely on statistical models and machine learning to identify promising talent, evaluate player potential, and build rosters strategically. At the same time, data-driven approaches enhance the fan experience through personalized content, interactive visualizations, and tailored marketing. With major tournaments like Euro 2024 showcasing these advancements, data analytics has proven its value not only in improving athletic performance but also in fostering deeper connections between clubs, players, and supporters.

Football is a multifaceted game where outcomes are shaped by a vast array of interacting factors. Data analytics allow teams to break down this complexity and uncover hidden patterns. By evaluating large datasets that include statistics, player movement, physiological metrics, and contextual in-game actions, clubs can identify trends, anticipate performance outcomes, and make evidence-based decisions. This analytical perspective enables both clubs and national teams to refine strategies, assess players accurately, and exploit opportunities that would otherwise remain invisible through traditional observation.

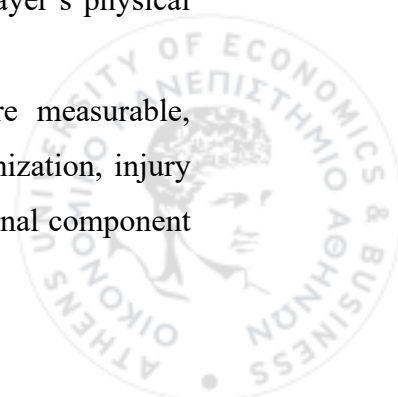


One core application of data analysis lies in training and match preparation. With the aid of modern tracking systems, teams collect precise data on sprint intensity, running distance, positional play, and other performance metrics. These indicators—such as maximum speed, number of sprints, passing accuracy, and tackle rate—are used to create individualized training programs tailored to player strengths and weaknesses. Moreover, by monitoring exertion and recovery indicators, clubs can identify injury risks early and adjust workloads, accordingly, maximizing player longevity and availability.

Data analytics also plays a pivotal role in scouting and player evaluation. Through machine learning algorithms and pattern recognition, scouts and analysts can assess long-term development trends and detect undervalued talent. Performance metrics from matches, physiological data, and even external sources like social media behavior can be synthesized to evaluate a player's potential fit within a team's philosophy. This analytical approach supports strategic roster development and helps identify players with not only technical skill but also mental attributes such as resilience and leadership (Schleutker, N. 2024).

On the tactical side, real-time data informs in-game decisions, enabling coaches to adapt formations, exploit opposing weaknesses, and make substitutions based on performance data. Opponent analysis using data visualization tools also allows teams to prepare more precisely for matches, uncovering tactical patterns and vulnerabilities. Data analytics has revolutionized football by changing how teams, coaches, and analysts operate. The analysis of extensive data sets enables teams to retrieve critical information about player performance, tactical decisions, and strategic development. Kalianiotis (2021) highlights that statistical analysis in football supports unbiased performance evaluation and informed decision-making. Moreover, innovations such as VAR and Goal-Line Technology have enhanced accuracy and transparency in the sport. Data analytics now serves broader purposes, including injury prevention and the creation of individualized training plans aimed at optimizing each player's physical condition and career longevity.

In conclusion, data analytics has transformed football into a more measurable, predictable, and optimized sport. Whether through performance optimization, injury prevention, tactical planning, or scouting, data has become a foundational component



of decision-making at all levels of the game. As illustrated in this research, xG modeling serves as one of the most illustrative examples of how complex football events can be quantified and analyzed for strategic advantage (Schleutker, N. 2024).

1.3 Applications of xG Prediction

The Expected Goals indicator has been used in different domains of football. In the pitch, coaches can use xG to assess the efficiency of their attacking strategies and where they can improve. For instance, a team that has many chances with high xG but scores few goals may require to give emphasis to train the finishing. Or we can identify which factors affect the xG indicator and in this way teams and staff can improve what they notice that is in low levels (for example, passes in the final third of pitch).Moreover in the field of scouting, xG assists in evaluating the players who are likely to create or exploit scoring opportunities even if they do not fully capture their performance. This enables teams to look for undervalued players with great potential (Mackenzie & Cushion, 2013). In the field of betting, xG is used to assess the performance of teams and forecast the results of matches. In this way betting companies will have better strategy for odds imputing and refreshing.



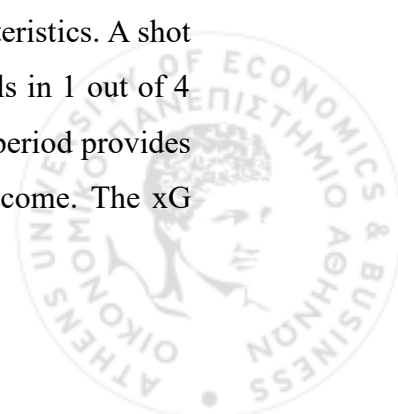
Chapter 2. Theoretical Background

2.1 Conceptual Approach to Expected Goals (xG)

The development of Expected Goals (xG) indicator became necessary because football needed a better system to measure attacking effectiveness objectively. Traditional statistics which track shots and possessions fail to deliver sufficient information about the quality of chances that teams or players create. The xG indicator addresses this information deficit by calculating the scoring probability of each final attempt based on phase characteristics. The xG concept recognizes that football opportunities differ in their value and goal-scoring probability. The modeling of goal-scoring attempt factors occurs through historical data analysis and statistical and machine learning applications. The goal is to develop a system which enables fair and knowledgeable odds evaluations between teams and players. This paper examines the xG concept together with its historical background and its connection to traditional statistics and factors that affect its calculation. This research provides a comprehensive evaluation of xG statistical analysis in football through its examination of value and limitations.

2.1.1 Definition of the Expected Goals indicator

The Expected Goals (xG) index is today as one of the modern and popular methods to evaluate both team and player attacking performance in football. The statistical model predicts goal probabilities from final shooting attempts based on their individual characteristics and environmental conditions. The xG system uses numbers ranging from 0 to 1 to quantify each shot's goal-scoring chances (Mackenzie & Cushion, 2013). The index is based on the idea that each shot has a different chance of scoring. The chance of goal conversion significantly increases when a player shoots from within the small area compared to taking a shot from outside this region. The calculation of xG depends on multiple variables including goal distance and shot angle, execution method, defender pressure and additional technical factors of the phase (Fernandez et al., 2021). Each shot receives an estimated xG value based on its characteristics. A shot with an xG rating of 0.25 indicates that similar situations produce goals in 1 out of 4 instances. The total xG value of a team's shots during a match or time-period provides an unbiased view of its attacking output regardless of the game's outcome. The xG

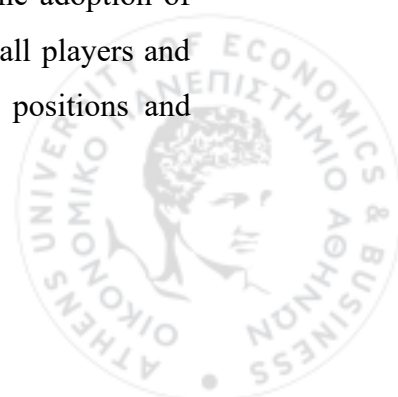


system evaluates the chances created by players but not their execution or quality. The gap between actual goals scored and expected goals scored (xG) helps identify if a team or player performs better or worse than expected from their created opportunities (Lucey et al., 2014). The Expected Goals indicator has gained universal recognition between academic circles and professional sports organizations because it presents a fresh approach to analyze attacking strategy effectiveness and player performance. The indicator serves multiple purposes in scouting reports and opponent analysis and result prediction and player evaluation beyond basic shot and attempt statistics.

2.1.2 Historical emergence and evolution of the xG methodology

The concept of Expected Goals (xG) emerged from the requirement to evaluate football offensive actions through objective qualitative methods. Until the late 2000s, the assessment of team performance depended on cumulative statistics such as shot counts and possession metrics without accounting for difficulty or quality levels in each phase. Number of final attempts failed to measure performance adequately because teams could take many non-threatening shots without becoming dangerous according to Anderson & Sally (2013). The first attempt to quantify chance quality emerged through unofficial applications developed for Opta and Infogol platforms during 2012. Early versions of xG models applied basic parameters that measured shot position relative to the goal from both distance and angle perspectives. The fast pace of technological development in data collection and analysis brought additional sophisticated elements for xG models such as attack characteristics and defensive player involvement and pass types (Mackenzie & Cushion, 2013).

During the years from 2014 to 2016 data analysis companies such as StatsBomb together with Opta started offering systematic data that enabled better xG model development for enhanced accuracy. During this time the research community dedicated itself to enhancing prediction accuracy by using machine learning techniques that included logistic regression and random forests (Fernandez et al., 2021). The evolution of xG methodology reached its critical milestone through the adoption of tracking data. New calculation methods for xG emerge from tracking all players and the ball throughout matches to include variables such as opponent positions and pressure and phase-wide dynamics (Lucey et al., 2014).

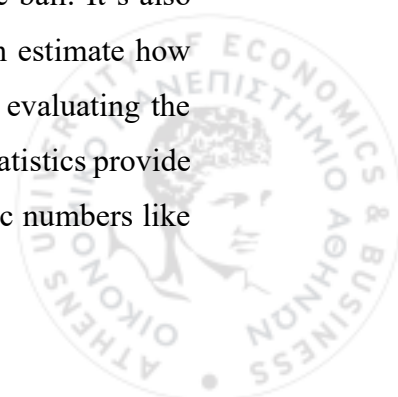


The acceptance of xG as a standard tool in football has occurred since the middle of the 2010s according to both academic researchers and professional football experts. Top-tier football clubs including Liverpool and Brentford use xG as their main performance analysis tool for making transfers and developing strategies (Anderson & Sally, 2013). The application of xG stands as a fundamental statistical evaluation method which organizations and research groups continue to enhance based on their specific requirements. Modern advancements in computing technology along with improved data sophistication enable future applications of the index to become more accurate and flexible.

2.1.3 Relationship of xG with traditional statistics

For many years, football performance was mostly measured using traditional stats like the number of shots, possession percentage, corners, or goal attempts. While these numbers offered a basic summary of what happened during a match, they had clear limitations mainly because they didn't capture the quality of chances created or the context in which those actions occurred (Anderson & Sally, 2013).

The Expected Goals (xG) metric was introduced to address this issue. Unlike simply counting shots, xG estimates the likelihood that a specific shot will result in a goal, based on important factors such as the distance and angle of the shot, the type of pass leading up to it, and the level of defensive pressure. As a result, two teams might both take the same number of shots in a match but have very different xG values. This difference provides a much clearer picture of which team created better scoring opportunities and how effective their attack really was (Mackenzie & Cushion, 2013). Furthermore, possession, another traditional football statistic, does not always tell the full story about a team's attacking efficiency or effectiveness. Some teams may control the ball for most of the game but still struggle to create meaningful or high-quality chances, something that only becomes clear through xG analysis. One of the key advantages of xG is that it separates the quality of chances from the overall volume of play, highlighting creativity and efficiency rather than just time on the ball. It's also important to mention related metrics like expected assists (xA), which estimate how likely a pass is to result in a goal. Like xG, xA builds on the idea of evaluating the probability behind actions, rather than just outcomes. These advanced statistics provide more insightful ways to assess individual players, moving beyond basic numbers like



goals or assists, which often depend heavily on team performance or match conditions (Fernandez et al., 2021)

The introduction of xG has changed how we understand performance and effectiveness in football. While traditional statistics focus only on outcomes like whether a goal was scored xG measures the likelihood of success. This makes it possible to detect whether players or teams are performing above or below expectations over time. In this way, xG doesn't just complement traditional football stats; it often improves them, leading to more evidence-based analysis and smarter decision-making in both academic research and professional football environments (Lucey et al., 2014).

2.2 Factors affecting the probability of a goal

Every team strives to score goals in football, but goals represent only a small fraction of total competitive actions during a match. Multiple parameters, which interact in complex ways, determine the probability of a shot resulting in a goal. The Expected Goals (xG) model analyzes these factors to measure the quality of each final attempt. The probability of scoring is influenced by several key variables, including shot location, type of final attempt, defender pressure, goalkeeper positioning, pass type, and the overall match context.

According to Rathke (2017), among these variables, the distance and angle from goal are particularly influential in determining scoring probability. Similarly, Hewitt and Karaküs (2023) highlight location, pass type, shot direction, and aerial duels prior to the shot as critical determinants of shot success. These findings emphasize the importance of spatial and contextual variables in the modeling process.

Overall, we understand that shot location which inherently captures aspects of distance and angle along with shot direction, represent some of the most impactful predictors of xG. The precise measurement of such parameters is essential for building reliable xG models and ensuring their effective application in professional football. The subsequent analysis examines each of these contributing factors in detail to reveal their individual and combined influence on opportunity creation and shot success.



2.2.1 Location of the shot

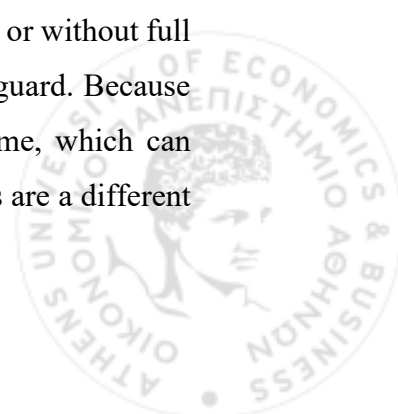
The location of a shot at the moment it's taken is one of the most important factors in determining the likelihood of scoring a goal. Research by Mackenzie and Cushion (2013) has shown that both the distance from the goal and the angle of the shot play a key role in influencing scoring probability. Players who take shots from central areas close to the goal have a much higher chance of scoring. In fact, shots taken from the small area often have an xG value of 0.5 or higher, meaning they result in a goal around half the time (Fernandez et al., 2021).

The angle of the shot is another critical factor. The chances of scoring drop significantly when a player shoots from a tight angle near the touchline, compared to shooting head-on. This is mainly due to the difficulty of aiming accurately and the limited view of the goal. Shot position also influences how goalkeeper and defenders can respond. When a shot is taken close to the goal, the goalkeeper has less time to react, which increases the likelihood of scoring (Lucey et al., 2014).

Modern football tactics often reflect this understanding. Teams like Manchester City under Pep Guardiola prioritize creating clear, high-quality chances in the box rather than relying on low-probability long-range shots (Anderson & Sally, 2013). In fact, evaluating xG and chance quality usually begins with analyzing the location of the shot. The calculation of xG based heavily on both the distance from the goal and the shooting angle to estimate how likely a given shot is to result in a goal.

2.2.2 Type of final effort

One of the key factors that influences whether a shot results in a goal is the nature of the final action. Several details come into play here the location of the shot, whether it was a kick or a header, if it was taken on the first touch or after controlling the ball, and the player's position and level of focus at the moment of the attempt (Mackenzie & Cushion, 2013). Shots taken with a player's stronger foot, the one they can strike with the most accuracy and power, tend to have a higher chance of success than those taken with their weaker foot. On the other hand, shots taken on the first touch, or without full control, may be less accurate, but they often catch the goalkeeper off guard. Because they happen quickly, these shots reduce the goalkeeper's reaction time, which can actually increase the likelihood of scoring (Lucey et al., 2014). Headers are a different



story. Even though they're usually taken from close range, headers typically have a lower xG compared to shots with the feet. That's because it's more difficult to direct the ball and generate power with the head. Research shows that goals from headers make up a smaller share of total goals, and their success depends heavily on the quality of the cross and the position of nearby defenders (Fernandez et al., 2021).

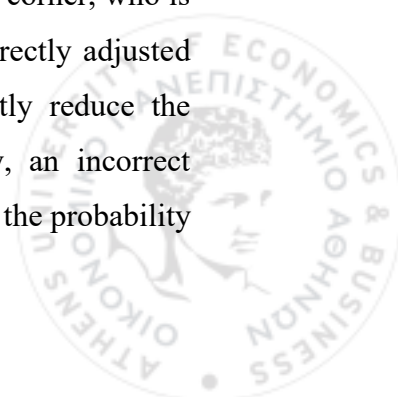
The type of play leading up to the shot also matters. A well-organized team move ending in a close-range pass (like a cutback) tends to produce higher-quality chances than a shot that follows a random rebound or long ball. In this way, how the chance is created directly impacts the xG value of the final shot (Anderson & Sally, 2013).

Finally, the player's balance and body position at the time of shooting play a major role in shot quality. A player who is stable and not under heavy pressure from defenders is more likely to hit an accurate shot. In contrast, attempts made while off-balance or closely marked are generally less effective and have a lower probability of resulting in a goal.

2.2.3 Pressure from defenders and goalkeeper position

The probability of a shot being converted into a goal is not only determined by the position or type of the final attempt but is also significantly influenced by the level of pressure produced by the defenders and the position of the goalkeeper at the time of the attempt. These dynamic factors affect the quality of the attempt and, consequently, the value of the expected goal (xG). The presence of defenders close to the player attempting the shot has a significant impact on the difficulty of the shot. The greater the pressure, the more the ball control, body balance and accuracy of the shot are reduced. According to research by the International Soccer Science & Performance Federation (2023), defensive pressure is one of the strongest factors in reducing the probability of scoring, with the probability falling by up to 40% when a player attempts a shot under strong pressure. On the contrary, in cases where the player shoots without an opponent close to him, the probability of scoring a goal increases dramatically.

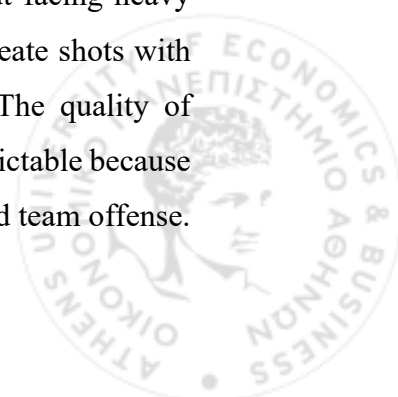
At the same time, the goalkeeper's position the moment of execution directly affects the success of the final attempt. A goalkeeper who correctly covers his corner, who is positioned at a short distance from the shooter (with the distance correctly adjusted based on the position of the shot), has the potential to significantly reduce the probability of scoring a goal (Sky Sports, 2023). On the contrary, an incorrect positioning, either too close to the goal line or out of position, increases the probability



of the shot finding the target. Modern advanced statistical analysis models, such as Expected Goals on Target (xGOT), attempt to calculate the real difficulty of a shot after taking into account not only the initial point of execution but also the point to which the ball was directed and the position of the goalkeeper (Fernandez et al., 2021). Thus, xGOT offers a more accurate measurement of the value of the final attempt, compared to the classic xG which only takes into account the execution conditions and not the final execution. The combination of defensive pressure and good goalkeeper positioning is a critical factor in reducing the probability of a goal. In phases where there is no pressure and the goalkeeper is out of an ideal position, the probability of success skyrockets. For this reason, top teams worldwide invest in positional and pressure analysis techniques, both to improve their defensive behavior and to identify weaknesses in their opponents (Anderson & Sally, 2013). In conclusion, defensive pressure and goalkeeper positioning are two of the most important dynamic factors that influence whether a shot will result in a goal. Incorporating these elements into xG prediction models significantly enhances the accuracy of statistical analyses and offers football professionals a more reliable tool for evaluating offensive and defensive strategies.

2.2.4 Type of assist or creation of a phase (e.g. cutback, through ball)

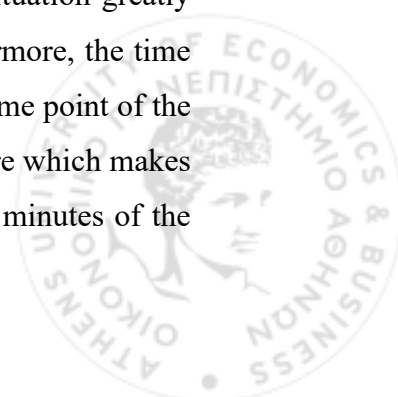
The probability of turning a shot into a goal depends heavily on the assist type together with the manner in which the passing play develops. The success rate of converting chances into goals becomes higher when chances are created under advantageous conditions but passes which stem from random occurrences or weak distribution tend to result in lower xG values (Lucey et al., 2014). Studies demonstrate that shots from short distances and central locations have higher average xG values than shots taken from distant angles (Fernandez et al., 2021). The football tactic of passing the ball toward the middle area after a lineout stands as one of the most successful methods to generate chances because it allows the player to find an advantageous position near the goal. Through passes that breach defensive lines generate exceptional scoring opportunities because they give shooters ample space to shoot without facing heavy defensive pressure. According to StatsBomb (2022), through passes create shots with substantially higher average xG values than static or long passes. The quality of opportunities generated from failed defensive actions becomes less predictable because these situations emerge from random occurrences instead of coordinated team offense.



The worth of these scoring chances strongly depends on the shooter's positioning and balance when they take the shot (Caley, 2015). The way a team generates their shots holds both practical value and strategic significance. Teams which devote resources to organized attacking strategies including combinations and cutbacks and vertical passes achieve better xG ratings that lead to more successful goal-scoring outcomes. Teams which depend primarily on set pieces and long balls generate fewer high-quality scoring chances (Mackenzie & Cushion, 2013). Modern football playmakers receive assessment based on their assist numbers and total expected assists (xA) which measures their chance creation value regardless of goal outcomes (Anderson & Sally, 2013). The quality of an opportunity together with the correct prediction of scoring probability depends entirely on the assist type and pass creation method. Every team can gain strategic benefit by optimally using these factors to enhance its offensive performance.

2.2.5 Match situation

The state of the game when the last attempt is made determines the chances of converting the shot into a goal. The general game situation, whether the move comes from open play, a counterattack or a set-piece, and the game situation, as well as the psychological state of the players, are all factors that are related to the difficulty and quality of an attempt. The statistical analysis has revealed that the opportunities taken through counter-attacks have a higher success rate as compared to those taken through set-pieces or set-pieces. In a counter-attack the opponent's defensive organization is usually disorganized, and this creates more space for the attacking player. Lucey et al. (2014) states that shots taken during counter-attacks have higher average xG values, due to the low defensive pressure and the good execution position. On the other hand, phases that begin with organized attacks (positional attacks) in well-organized defenses tend to have lower xG values, since the defending team is better positioned, limits distance and closes the corridors. In the same way, attempts from set pieces (corners, free kicks) are much more difficult to score from. Although these phases generate a lot of final attempts, the defenders' density and the static nature of the situation greatly diminish the quality of the chances (Anderson & Sally, 2013). Furthermore, the time factor is very important. Many studies have shown that the closer the time point of the match is to the end of the match, the more the players are under pressure which makes them less accurate in their actions. The last attempts made in the last minutes of the

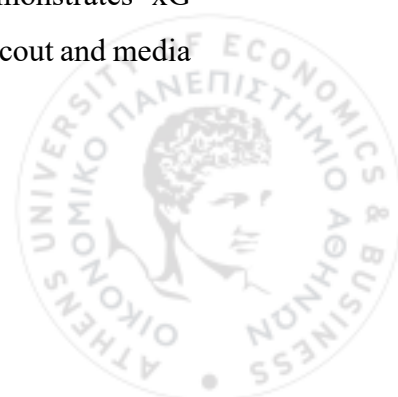


match, especially if the team is losing, are often made with a lot of pressure and stress which makes it difficult to be effective even in high xG phases. The state of the score is also another factor that should be considered. The teams that are leading will slow down and play more defensively while the teams that are chasing the lead will increase the tempo and take more risks. This change in the dynamics of the game affects both the number and quality of chances created, as StatsBomb (2022) reports, highlighting the importance of “contextualized analytics” — analysis that takes into account the full context of the match.

2.3 Example of use of xG in teams and analysts

Expected Goals (xG) serves as a statistical tool which extends beyond academic and research environments. The tool has established significant real-world applications throughout professional football which shapes practices and decision-making at different operational levels. The modern performance analysis includes xG as an essential element which technical directors and coaches together with data analysts and scouts and journalists use. The value of xG emerges from its ability to provide advanced performance insights which extend past basic statistics about goals and shots and assists. Through xG analysis teams gain an unbiased measure of their created and conceded chances which enables them to evaluate tactical strategies and player effectiveness through data instead of visual observations.

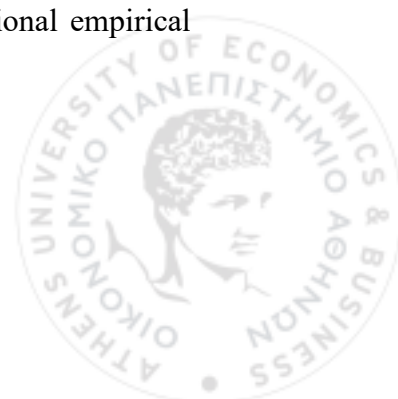
The development of xG has prompted numerous professional teams to establish dedicated data analysis departments which track xG and xA (expected assists) and additional predictive performance models. The same analytical tools help scouting operations and transfer evaluations to detect players who generate or perform valuable scoring opportunities regardless of their conventional statistical performance. The xG metric serves independent football analysts and journalists to understand match results through the comparison between actual goals and expected goals. The gap between actual goals and expected goals helps identify team luck levels and assesses their performance relative to chance quality. The following section demonstrates xG applications through professional team examples as well as analyst and scout and media examples to show the indicator's extensive use in current football.



2.3.1 Use by professional teams

The quick advancements in data analysis methods enable professional football teams to incorporate Expected Goals (xG) indicators into their decision-making process and evaluation methods. High-level clubs use xG beyond basic performance analysis to build it into their competitive strategy and transfer policies. Brentford FC represents one of the most distinctive examples of xG application since they started using data analytics beginning in the mid-2010s. The team's manager Matthew Benham who studied statistical models for betting, created an entire scouting and tactical system using advanced metrics including xG and xA (expected assists). The team achieved Premier League promotion alongside Championship success through its exclusive use of performance data according to Guillermo Martinez (2018). The Danish club Midtjylland which Benham owns implements analytical tools like xG to help make both tactical selection and transfer decisions. The team uses probability analysis to evaluate chances and goals as part of their systematic evaluation process. The system enables them to find players who generate worth even though their traditional statistics are not outstanding (Anderson & Sally, 2013). The English top league team Liverpool FC demonstrates how statistical analysis through xG has become part of their strategic approach. Ian Graham leads the analytics team at Liverpool which employs xG analysis to evaluate both attacking line efficiency and defensive team performance. Through these analytical methods Liverpool discovered that Mohamed Salah possessed high-quality chance creation abilities which led to his undervalued status before joining Anfield. xG serves professional teams for both short-term match analysis and developing their long-term strategic plans. Expected goals data enables teams to make transfer choices and arrange their starting lineups and develop training programs for final moves based on statistical patterns. The analysis departments across numerous teams now function as "statistical laboratories" which deliver immediate recommendations to both coaches and team executives. xG evolved from its theoretical statistical roots into a practical strategy instrument for top professional clubs. The analysis of such indicators now becomes essential for teams at all levels because evidence-based decisions produce competitive advantages over traditional empirical observation.

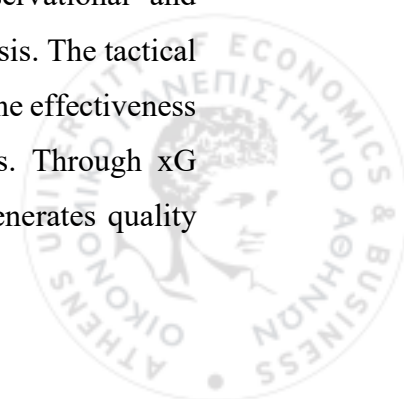
2.3.2 Use by analysts, scouts, sports journalists



Professional football teams no longer hold the exclusive right to Expected Goals (xG) usage which extends into widespread application beyond their facilities. The xG tool has become essential for independent football analysts together with scouts and statisticians and sports journalists to enhance interpretive accuracy and document team and player performance. Modern performance analysis requires xG because this metric uncovers the actual processes occurring during match results. The xG index serves data analysts who work independently or for specialized media such as StatsBomb FBref or The Analyst to evaluate both team stability and long-term viability. The low xG rating combined with many scored goals indicates that the team exceeds its potential for long-term success. Teams who possess high xG but score few goals either face bad luck or fail to convert their chances effectively. The xG indicator serves scouts alongside xA (expected assists) to measure players who generate high-quality scoring opportunities. Through this method they can detect players who make major contributions to chance creation regardless of their infrequent appearance in highlight reels or standard statistics. Sports journalists play an essential role by using indicators like xG in their analyses to break free from basic "won/lost" superficial approaches. Through xG timelines and heatmaps combined with graphs they demonstrate to viewers a detailed match overview which shows both quality and quantity of attacks. The website Understat.com functions as a widely recognized platform for analysts and journalists to analyze team performances through xG and xG Against (xGA) statistics (Understat, 2023). The introduction of xG statistics now appears on both betting websites and television sports broadcasts which enables public viewership of advanced technical metrics. The increased availability of quantitative performance analysis has both brought the sports fan base closer to data-based insights and made them question score-based interpretations.

2.3.3 Impact on tactical analysis and player selection

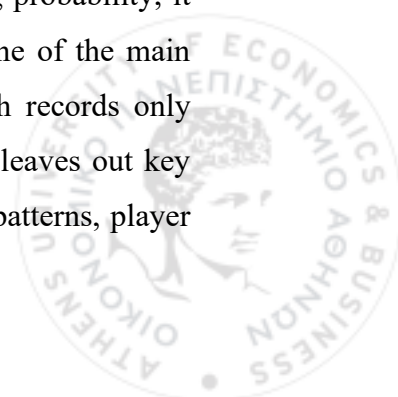
Expected Goals (xG) introduced major transformations which affect performance assessments of teams and players and basic tactical analysis and roster building processes. The basis for making decisions has shifted from observational and experiential methods to combine empirical data with quantitative analysis. The tactical analysis field benefits from xG as an essential tool which helps decode the effectiveness of particular formations and attacking plans and defensive methods. Through xG analysis coaches and analysts can determine if their strategic plan generates quality



chances or not. Two teams that produce equal numbers of final attempts will show distinct xG levels which demonstrates the actual variation between phase quality (Anderson & Sally, 2013). The analysis reveals which tactics are strong or weak based on metrics including cutbacks frequency and vertical pass finals and second phase exploitation. Examining match xG differences allows teams to modify their playing approach. A team that generates low xG chances should consider adopting new tactical approaches through wing overlaps and reducing their dependence on long-distance shots. The annual Stats Perform analysis (2022) demonstrates that teams which modify their attacking strategy through xG metrics achieve better scoring results and improved positional harmony in the attacking area. xG analysis greatly changed how football players are evaluated and chosen. Footballers now receive evaluation based on their goal and assist numbers as well as their ability to generate high-quality scoring opportunities. Footballers who maintain high xG values during each 90-minute match retain their value despite low goal conversion rates because their efficiency levels are expected to stabilize across time. The evaluation of players through xG over- and under-performance measurements provides significant insights into long-term performance potential. The xG statistics show that a goal-scoring striker who outperforms their expected goals might be experiencing a hot streak yet lack long-term potential. A player who generates or executes scoring opportunities but fails to score could be valuable for teams with suitable strategic adjustments. The xG indicator introduced a fresh element which teams apply to their tactical planning and transfer operations. The quantitative assessment of action quality through this indicator serves as a fundamental instrument for making strategic and informed decisions.

2.4 Problems and Limitations in xG Prediction

Although the Expected Goals (xG) metric has become increasingly popular and useful in football analysis, it still has several important limitations. xG models are built on basic assumptions and are affected by the quality and type of data available. These limitations can reduce both the accuracy of predictions and how easily the results can be interpreted. While xG helps us understand shot quality and scoring probability, it doesn't fully capture the dynamic and complex nature of football. One of the main weaknesses is that most xG models rely on event-based data, which records only specific actions like shots, passes, or tackles. This type of data often leaves out key elements such as player positioning, defensive shape, ball movement patterns, player

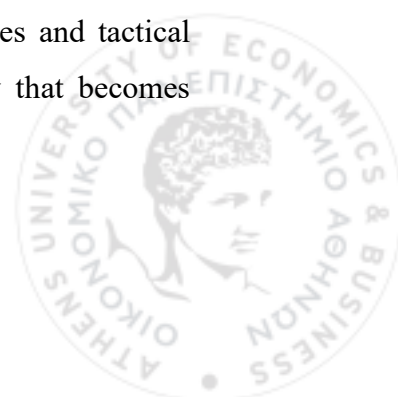


fatigue, or the psychological pressure players may face in high-stakes moments. Traditional models also struggle to consider individual player skill levels, mental states, and tactical differences between teams. They tend to treat all players and situations the same, which limits how much insight the model can truly offer. Because of this, xG should be seen as a helpful analytical tool—but not as a complete or perfect measure of performance.

The next two sections will explore the main methodological and theoretical limitations of xG prediction, focusing on the gaps in event-based data and the structural challenges of using traditional models to understand football in real game situations.

2.4.1 Subjectivity and Loss of Information in Event-Based Data

Public xG models depend mainly on event-based data which includes information about what happened during matches such as shot locations and types and assist origins and block status. The features used for analysis in these systems have restricted capabilities because they fail to deliver enough specific information about event causes and game-wide influences (Mackenzie & Cushion, 2013). Human coders manually tag event data which brings both personal interpretation and possible biases into the event labeling process. The exact position of a shot depends on video analysis while analysts might classify passes differently through terms like “through ball” or “cutback.” The inconsistent nature of these data points leads to model accuracy issues which in turn reduces the reliability of xG output when comparing data from different providers or competitions (Lucey et al., 2014). The main drawback of event-based datasets is their inability to capture how the game unfolds through space and time. The system fails to monitor how players move or how defensive formations change or how teams transition between different areas of the field in real time. Real match situations result in vastly different success probabilities between two shots that share identical location and angle characteristics. Both shots receive identical xG values even though one occurs during an unguarded fast break and the other occurs in a crowded defensive area with tight marking (Fernandez et al., 2021). The model's ability to deliver meaningful insights remains restricted because of this simplification especially for coaches and tactical analysts. The absence of context in xG creates a performance view that becomes misleading when used for independent analysis.



2.4.2 Limitations in Models: Does Not Capture the Context of the Game

Analysis of xG faces a critical restriction because it fails to understand match contexts which play a significant role in determining action results. Advanced models use variables like shot location and angle yet fail to measure in-play dynamics including player movement before the shot body orientation fatigue levels team strategy and defensive disruption. The analysis of context reveals its deepest implications during tactical transitions such as counter-attacks and high-stakes situations where psychological pressure affects execution. The probability of success differs between a friendly match shot with no pressure and the same shot taken during the 90th minute of a cup final. Most xG models apply equivalent calculations to situations which Sky Sports (2023) identifies as distinct. xG models lack the capability to track live defensive positions and goalkeeper locations during shots. The tagging system of StatsBomb includes basic categories like “under pressure” and “fast break” but these categories lack the specific tracking data precision (StatsBomb, 2022). The absence of tracking data leads analysts to overlook crucial small variations in shot quality. The current research in academia and commercial sectors works toward enhancing context-specific xG modeling. Expected Possession Value (EPV) and Expected Threat (xT) represent two models which predict scoring chances across entire possession sequences instead of focusing on single final shots (Fernandez et al., 2021). These approaches need high-resolution tracking data which teams and analysts without access to this luxury cannot use. The xG evaluation tool excels at assessing both shot quality and team attacking performance yet represents only a limited view of game dynamics. A comprehensive contextual model needs tracking information together with psychological data and situational insight which standard xG implementations currently cannot support.



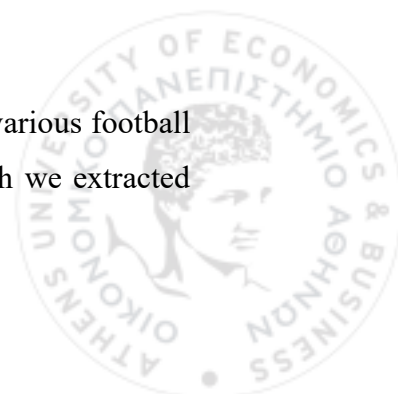
Chapter 3. Methodology and Data Sources

3.1 Data Source

StatsBomb Open Data provides football match information which can be accessed for research purposes through their open data platform StatsBomb Open Data. The data collection stands as one of the most dependable football statistical datasets since it offers detailed event-based information across multiple competitions during different periods. This research employs data from the Premier League season that occurred between 2015 and 2016. The researcher chose this season to create an integrated dataset because it included enough matches along with complete event recording and appropriate representation of top division English competition and match intensity. The detailed match data available through StatsBomb contains all event types including shots passes, ball contacts, defensive actions, pressure phases, action locations and time stamps. The phases include specific characteristics that describe pressure status, distances from the goalkeeper and the goal, angles of the shot, as well as contextual details such as the number of defenders in the shooting cone and whether the goalkeeper was positioned within it.. The elements serve to evaluate offensive phase quality and calculate Expected Goals (xG) (Fernandez et al., 2021). The researcher obtained the data from StatsBombR which serves as an R programming language package developed by StatsBomb for data retrieval. The StatsBombR package enables users to download and sort data automatically from the GitHub public repository maintained by StatsBomb (StatsBombR, 2023). StatsBombR offers researchers a standardized data framework and seamless integration with additional statistical software available in R. Research articles worldwide use StatsBomb data extensively to build xG models because this dataset provides accurate recording capabilities alongside detailed meta-information compared to other public datasets which lack detailed data. The StatsBomb Open Data project provides open access to its documentation and datasets which makes it perfect for academic research and open-source initiatives.

3.2 Data Description

The StatsBomb Open Data project provides detailed match data from various football competitions. The initial dataset includes 72 competitions, from which we extracted



only the Premier League data for the 2015–2016 season. Overall, the dataset contains approximately 3,500 matches with around 1.5 million recorded events collected through event-based tracking. To narrow the focus, we filtered the data to include only shot events along with their relevant variables. After performing data transformations and cleaning, the final dataset consists of approximately 9,800 shot events across 35 variables. Every recorded event in the dataset includes time information alongside action type details, field location coordinates, team and player IDs and StatsBomb metadata. The research focuses on collecting data about final attempts above all else. The data includes the shot execution position together with the shot angle and body part used and assist type and pressure status and phase outcome (goal, save, out). These details serve as fundamental inputs for building Expected Goals (xG) prediction models because they evaluate the excellence of each final effort (Fernandez et al., 2021). The dataset contains information about passes as well as defensive actions such as blocks, interceptions and duels, fouls and static match phases like corners and penalties. The recording of pressure events enables researchers to track essential details about offensive or defensive match pressure points. The data uses a 120x80 reference system to express the field coordinates, where “location.x” and “location.y” variables represent the horizontal and vertical positions of each event on the pitch. In the dataset, these variables specifically describe the location from which a shot was taken, allowing spatial analysis of shooting behavior.

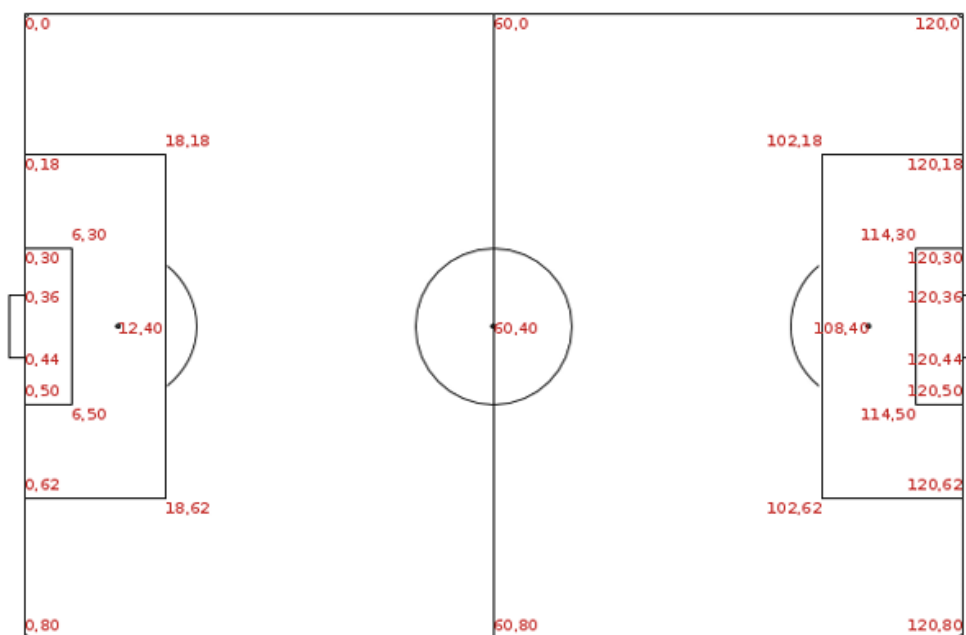
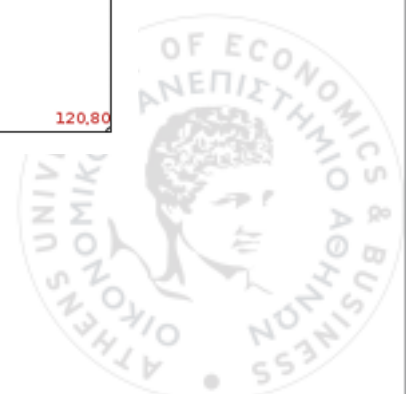


Figure 1: X-Y Coordinate System of the Pitch



The system supports both geometric analysis and spatial visualization of match events, which is helpful for exploratory data analysis (EDA) and building predictive models. The dataset includes information about both teams and individual players, such as unique IDs, names, and playing positions, along with various match-related details. One useful feature of the dataset is that it doesn't just record where each shot begins it also shows where the shot ends, using the variable "shot.end_location." This variable includes three values: x, y, and z. These represent the horizontal position along the length of the pitch (x), the side-to-side position across the pitch (y), and the height of the ball from the ground (z).

The figure below illustrates how this 3D coordinate system is applied to represent where shots end particularly those that reach or cross the goal line. For example:

- A coordinate like **(120, 36, 0)** represents a shot ending at the bottom-left corner of the goal.
- A coordinate like **(120, 44, 2.67)** corresponds to a shot reaching the top-right corner.

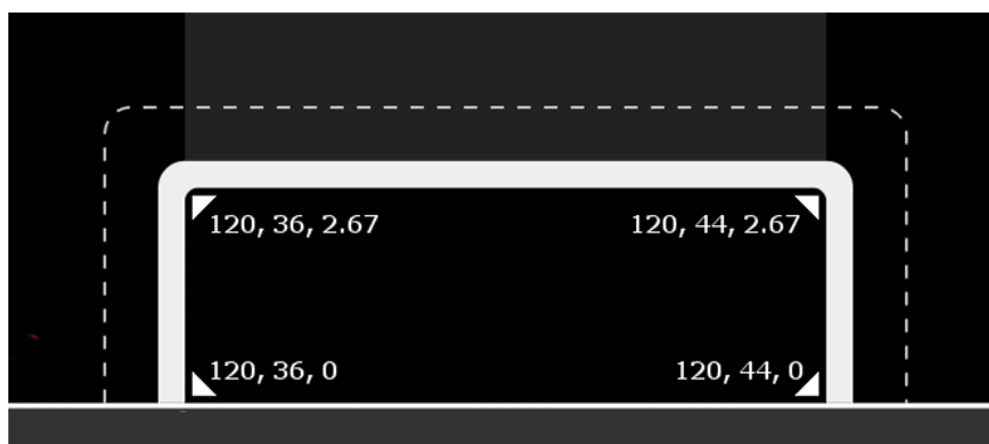


Figure 2: 3D Shot Endpoint Coordinates at the Goal Line

Finally, the image below depicts the positions of players across the football pitch, each labeled with a numeric and abbreviation code corresponding to their tactical role. This standardized layout helps categorize events and actions based on player positions in the dataset.

The positions are separated across defensive, midfield, and attacking roles, covering multiple formations and tactical setups. Below is a breakdown of the key positional abbreviations:

- **Goalkeeper (GK)** – 1
- **Defenders:**
 - CB: Centre Back (4), LCB/RCB: Left/Right Centre Back (5/3)
 - LB/RB: Left/Right Back (6/2)
 - LWB/RWB: Left/Right Wing Back (8/7)
- **Midfielders:**
 - CDM: Central Defensive Midfielder (10)
 - LDM/RDM: Left/Right Defensive Midfielder (11/9)
 - CM: Central Midfielder (14)
 - LCM/RCM: Left/Right Central Midfielder (15/13)
 - LM/RM: Left/Right Midfielder (16/12)
 - CAM: Central Attacking Midfielder (19)
 - LAM/RAM: Left/Right Attacking Midfielder (20/18)
- **Forwards:**
 - LW/RW: Left/Right Winger (21/17)
 - LCF/RCF: Left/Right Centre Forward (24/22)
 - ST: Striker (23)
 - SS: Second Striker (25)



Figure 3: Football Player Position Map with Role Codes

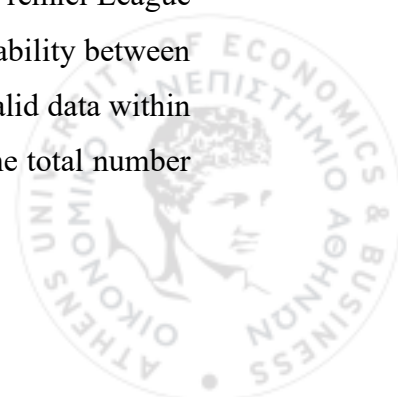


The lineups file contains information about starting lineups and changes for each team during each match while the matches file holds basic match data (date, venue, score, duration). Event data can be connected to match conditions through this dataset for comprehensive analysis across various levels. The JSON format of the dataset becomes structured data frames through the “StatsBombR” package in R for improved analysis. The detailed characteristics of this dataset support the development of statistical and machine learning models for match phase prediction and interpretation.

In summary, the dataset extracted from the StatsBomb Open Data project offers a rich and detailed event-based representation of football matches, specifically filtered to include shot events from the 2015–2016 Premier League season. With approximately 9,800 shot observations and 35 variables, the final dataset includes spatial information using a standardized 120x80 coordinate system, allowing accurate tracking of both shot initiation and end locations. It further incorporates contextual data such as player positioning, pressure phases, and the shot cone—an area representing the spatial field between the shooter and the goal, impacted by defenders and the goalkeeper. This granular level of detail supports in-depth spatial and tactical analysis, forming a solid foundation for performance evaluation and predictive modeling in football analytics

3.3 Data Cleaning and Processing

Any statistical analysis requires thorough data cleaning and processing especially when working with large multidimensional sets such as those found in this work. The “StatsBombR” package produced event-based data that contained many match events which varied in their characteristics and depth. Before starting the xG model development process, Wickham (2014) recommends that researchers must carefully clean and prepare their data. The first step was to filter the data to retain only events related to final attempts (shots). Using the event types provided by StatsBomb, we excluded unrelated actions such as substitutions, fouls, and other non-shot events that do not contribute directly to goal-scoring opportunities. This ensured that the dataset focused only on shot-specific variables relevant to expected goal (xG) modeling. We then restricted the dataset to only include matches from the 2015–2016 Premier League season. This decision provided a consistent dataset and minimized variability between different competitions. Following this, we addressed incomplete or invalid data within the dataset. Variables with more than 75% missing values relative to the total number



of observations were excluded from the analysis to ensure data quality and reduce noise from sparsely populated features.

In line with best practices in xG modeling, penalty shots were removed from the dataset. Penalties represent a special case in football with a fixed location and typically high probability of scoring (xG around 0.77). Including them would skew the model by artificially inflating prediction accuracy due to their predictability and distinct context. Additionally, 13 observations were removed because they had substantial missing information across several key variables. These observations only included the goal outcome (1 or 0), without sufficient contextual features. Given that they represent less than 0.1% of the dataset (13 out of approximately 9,800), their exclusion was deemed statistically insignificant and necessary for maintaining data integrity.

To handle other moderate levels of missing data, imputation techniques were employed. For instance, the variable “shot.end_location.z” — representing the height of the shot endpoint — had several missing values. These were filled using interpolation, providing context-aware estimates while preserving the three-dimensional spatial characteristics of each shot. Furthermore, variables in the dataset were systematically categorized as either numeric or categorical (factor) to support appropriate modeling techniques. Several character variables that describe qualitative aspects — such as “position.name” and “shot.technique.name” — were converted to factors, enabling their correct statistical treatment in logistic regression models. Binary indicators such as whether the shot was taken under pressure or followed a cutback pass were also included to enhance the model's ability to capture situational context. The R packages “dplyr” and “lubridate” were used to restructure and manage the dataset efficiently. After preprocessing, the cleaned dataset consisted of approximately 9,800 shots with more than 30 features describing spatial, temporal, and situational conditions around each shot attempt. This dataset became the foundation for exploratory data analysis (EDA) and the development of the expected goals (xG) prediction models. The accuracy and credibility of the results rely heavily on the rigor of the data cleaning and preparation process, as even small inconsistencies can distort outcomes in complex statistical analyses.



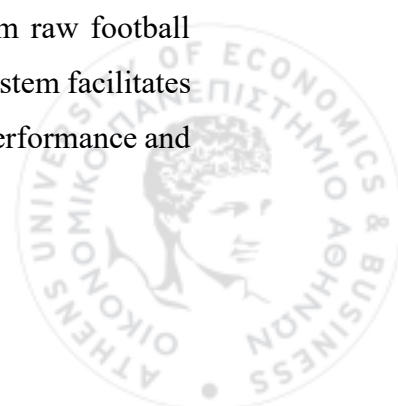
3.4 Tools and R Packages Used

The analysis and processing of the data in this project was conducted entirely within the R programming language using a set of specialized packages that help with data management analysis and model creation. R is one of the most well-known languages for statistical analysis and data science, due to its open architecture, strong community support and large package library (Wickham & Grolemund, 2017). The main tool used was “StatsBombR” package, which allows for direct connection and retrieval of data from the StatsBomb Open Data platform. The package gives functions for loading match data, organizing events and converting JSON data into structured data frames. The use of “StatsBombR” ensured the homogeneity and accuracy of the initial data, forming the foundation for all subsequent analytical procedures.

Key packages such as “dplyr”, “tidyr”, and “purrr” from the “tidyverse” ecosystem were used extensively for data manipulation, cleaning, and reshaping. These tools enabled efficient filtering, transformation, and handling of missing data within the raw event-level football dataset, allowing for the preparation of a clean and structured dataset suitable for analysis.

For exploratory data analysis (EDA) and visualization, “ggplot2” package was used, which provides a flexible grammar system for creating graphs and diagrams (Wickham, 2016). The “ggplot2” was particularly useful for visually analyzing the xG distribution, visualizing the location of shots on the court and studying the relationship between variables. Statistical evaluation was performed using packages like “sjmisc”, “nortest”, and “aod”, which enabled univariate and bivariate analyses, as well as normality and distribution testing. The modeling phase incorporated both classical logistic regression and regularized methods like LASSO, using packages such as “glmnet”, “caret”, and “pROC” for model fitting, validation, and performance metrics. Each model Full, AIC-optimized, BIC-optimized, and LASSO was evaluated using accuracy, precision, recall, F1-score, ROC curves, and McFadden’s R^2 , providing a comprehensive comparison of predictive performance.

This R-based workflow demonstrated the full analytical pipeline, from raw football event data to interpretable statistical models. It highlights how R's ecosystem facilitates both statistical rigor and flexibility, enabling deep insights into player performance and team strategies using structured data-driven approaches.



Chapter 4. Exploratory Data Analysis (EDA)

The Exploratory Data Analysis (EDA) phase is an important step for understanding the structure and patterns in the dataset. In this chapter, we start by showing basic descriptive statistics at both the team and player levels, giving an overview of key performance measures such as goals, expected goals (xG), and efficiency.

After that, we take a closer look at the variables in the dataset. Using statistical methods and visual tools, we explore how the data is distributed, how different variables relate to each other, and whether any strong correlations exist. This process helps us identify trends, find unusual data points, and gain insights that will guide the modeling stage and help improve the accuracy of predictions.

4.1 Statistics by Team

The first stage of Exploratory Data Analysis (EDA) is to collect key team statistics for the 2015/16 Premier League and to understand the basic structure of the league. The main objective is to identify teams that have been successful (or not) in terms of expected results.

The following key indicators were calculated for each team:

- Total number of shots taken in the season,
- Total number of goals scored,
- Total Expected Goals (xG), i.e. the sum of the scoring probability for all shots,
- Goal Difference – xG (Over/Underperformance), which shows whether a team scored more or less than it “should” have scored given the quality of its chances.

The results of the analysis indicated that teams like Manchester City, West Ham and Everton had the highest total xG, which could be attributed to their attacking approach and ability to generate chances. On the other hand, teams that finished at the bottom of



the table like Aston Villa or Watford had very low xG, which could mean they were not very good at creating chances.

An interesting finding was observed from the Goal – xG difference analysis:

- This season's champions, Leicester City, had a positive $G - xG$ difference, which means they scored more goals than expected from their chances. This was due to the fact that Jamie Vardy (as we will see in the next chapter at statistics per player) and other players were very efficient, but also due to their ability to perform at important moments.
- On the other hand, teams such as Arsenal had a negative difference, which means they underachieved in terms of the quality of chances they created.

To make a fair comparison between teams that played a different number of matches or had a different style of play, average shots per match and xG per shot were calculated. For example, teams that played a counter-attacking style of play would have fewer shots, but a higher average xG per shot, compared to teams that played a possession-based style of play, which would have more shots but a lower average xG. This is a team-level analysis which shows that it is not the number of shots that matters, but the quality of chances (xG) that a team gets. At the same time, the difference between goals and xG is an indicator of either excellent execution or luck, which can explain why the team deviated from the statistically expected results (Fernandez et al., 2021).

Finally, team level EDA also revealed the best and the most effective (or lucky) teams in the 2015–2016 Premier League, which gives important clues for the next stages of the work.



team.name	Total_Goals	Total_xG	Goal_minus_xG	Efficiency_Percentage
Arsenal	61	63.6	-2.57	96.0
Tottenham Hotspur	63	58.4	4.56	108.0
Manchester City	66	58.5	7.15	113.0
Liverpool	60	56.8	3.19	106.0
Leicester City	58	57.0	1.0	101.0
West Ham United	60	51.4	8.61	117.0
Southampton	60	51.8	8.2	116.0
Chelsea	59	51.4	8.61	117.0
Everton	53	48.3	4.75	110.0
Sunderland	41	40.1	0.863	102.0
Manchester United	43	39.8	3.15	108.0
Watford	30	38.6	-8.6	77.7
Newcastle United	44	38.3	1.69	104.0
Crystal Palace	32	38.2	-6.2	83.8
Swansea City	36	38.2	-2.15	94.4
AFC Bournemouth	40	37.2	2.78	107.0
Norwich City	36	37.2	-1.17	96.9
Stoke City	35	36.0	-1.99	94.5
West Bromwich Albion	31	34.7	-3.71	89.3
Aston Villa	23	28.0	-5.01	82.1

Table 1: Team Performance Comparison: Actual Goals vs Expected Goals (xG) and Efficiency (%) – Premier League 2015/2016

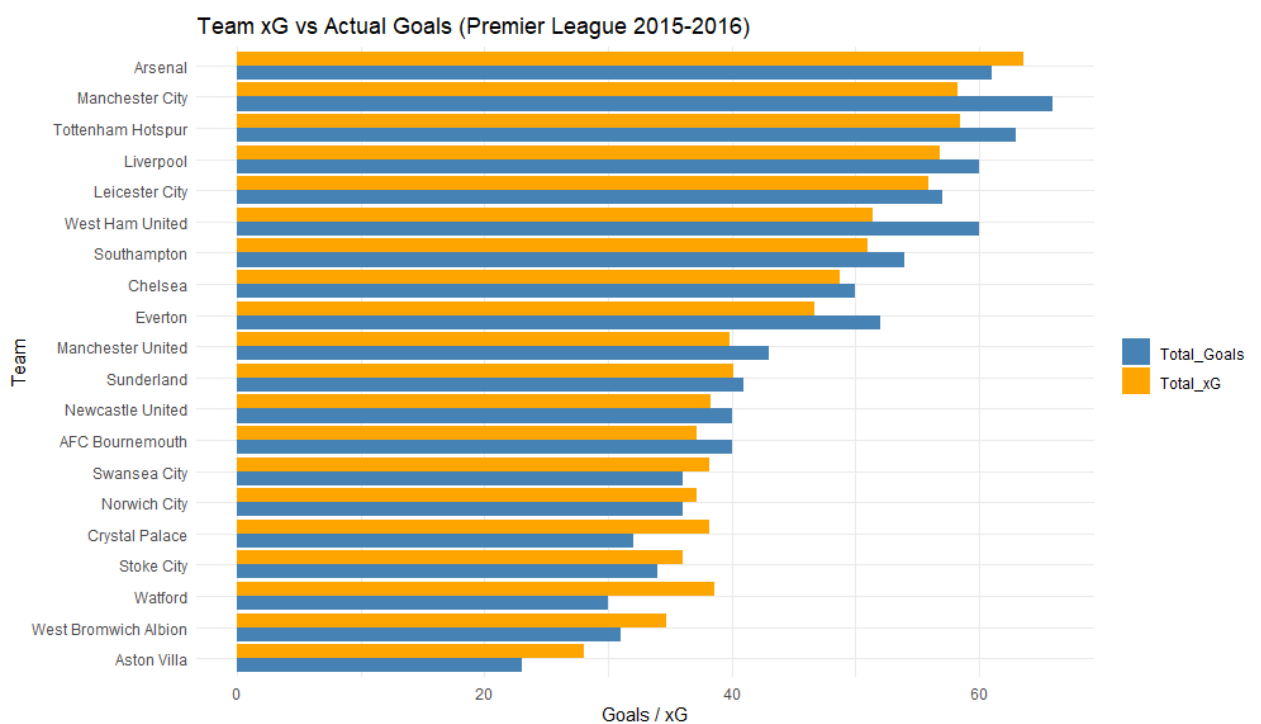


Figure 1: Horizontal Bar Chart of Actual Goals vs Expected Goals (xG) per Team – Premier League 2015/2016



4.2 Statistics by Player

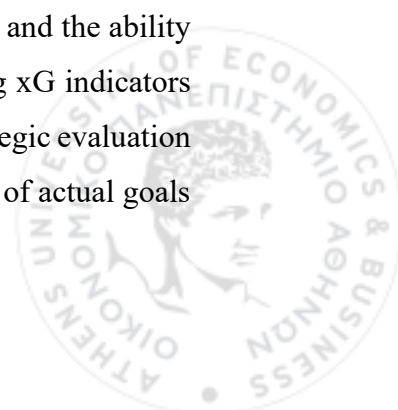
During the second stage of Exploratory Data Analysis (EDA) the examination moves from team assessment to individual player evaluation. The research focuses on determining player performance through key statistics which include goal counts and expected goals (xG) and shots and efficiency measures.

The analysis calculated the following metrics for each player during the 2015–2016 Premier League season:

- Total number of goals scored,
- Total number of shots taken,
- Expected Goals (xG) which resulted from their shot attempts,
- Goal Difference – xG (Overperformance/Underperformance),
- Efficiency (%): Goals / xG .

The analysis concentrated on players who appeared in a large number of matches and took at least 20 final shots throughout the season to guarantee reliable statistical results. The Premier League top goal scorers Harry Kane, Sergio Agüero and Jamie Vardy occupied the highest positions in goal and xG totals during the 2015-2016 season according to Premier League data (2016). Special attention needs to be paid when evaluating the goal numbers against xG metrics. The success rate (Goals / xG) allows the evaluation of a player's "natural" efficiency in scoring goals. Efficient goal scorers achieve more goals than xG expectations which proves vital for competitive evaluation and scouting operations. Scatter plots and ranking plots that connect xG to goals help analysts understand player contributions and detect players who excel beyond standard statistics.

Performance evaluation must consider both the number of goals scored and the ability to create quality chances and convert them into successful shots. Using xG indicators within individual assessments creates fairer and more objective and strategic evaluation methods. The Efficiency metric was calculated by dividing the number of actual goals



by expected goals (xG) and multiplying the result by 100 to express it as a percentage. This metric helps evaluate whether a player is scoring more (or fewer) goals than expected based on the quality of their chances. An efficiency over 100 means the player is outperforming their xG, while values below 100 indicate underperformance.

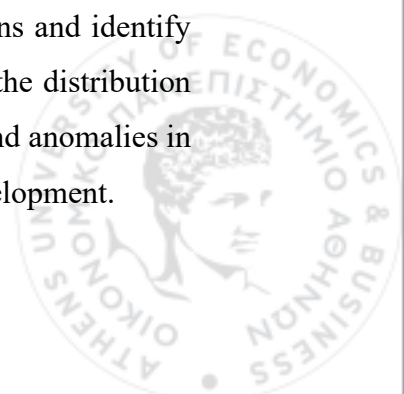
Player Performance Table					
player.name	Goals	xG	Shots	Difference	Efficiency
Harry Kane	20	17.72	153	2.28	112.88
Sergio Leonel Agüero del Castillo	20	13.76	113	6.24	145.40
Jamie Vardy	19	17.44	111	1.56	108.94
Romelu Lukaku Menama	17	16.59	114	0.41	102.47
Odion Jude Ighalo	15	16.92	111	-1.92	88.66
Olivier Giroud	15	12.23	102	2.77	122.61
Jermain Defoe	14	12.43	87	1.57	112.66
Alexis Alejandro Sánchez Sánchez	13	11.26	111	1.74	115.42
Riyad Mahrez	13	7.46	81	5.54	174.20
André Ayew Pelé	12	7.59	69	4.41	158.04

Table 2: Top 10 Scorers of Premier League 2015/2016

4.3 Univariate Analysis of Variables

Univariate analysis focuses on exploring and understanding the individual characteristics of each variable in the dataset. This type of analysis is crucial for gaining insights into the distribution, central tendency, and spread of both numeric and categorical variables before building predictive models. In this section, we apply various statistical and graphical techniques such as histograms, boxplots, frequency tables, and summary statistics to examine each variable independently.

The variables are grouped into two categories: numeric variables and factor variables. The main aim in this part is to indicate trends, and characteristics of the variables through a collection of summary statistics and visualizations that exceed information. For numeric variables, we assess the shape of distributions and identify potential outliers, while for categorical (factor) variables, we analyze the distribution of levels and their frequencies. These analyses help to reveal patterns and anomalies in the data and guide subsequent steps in feature selection and model development.



In the initial stages of our Exploratory Data Analysis (EDA), we transformed the 'Goal' column into a binary variable, assigning a value of 1 to indicate a Goal made and 0 to denote a shot that was not a goal. This binary representation facilitates our analysis by simplifying the interpretation of cancellation behavior within the dataset.

Goal					
<i>val</i>	<i>label</i>	<i>frq</i>	<i>raw.prc</i>	<i>valid.prc</i>	<i>cum.prc</i>
0		8895	90.73	90.73	90.73
1		909	9.27	9.27	100.00
NA	NA	0	0.00	NA	NA
<i>total N=9804 · valid N=9804 · $\bar{x}$=0.09 · σ=0.29</i>					

Table 3: Frequency table of Goal variable

The frequency distribution table for the Goal variable reveals a significant class imbalance in the dataset. Out of a total of 9,804 shot events, only 909 (9.27%) resulted in a goal, while the remaining 8,895 (90.73%) did not. This imbalance highlights the challenge in predicting goal outcomes, as goals are relatively rare compared to missed or saved attempts. The mean value of the Goal variable is 0.09 with a standard deviation of 0.29, further confirming the skewed nature of this binary variable. Addressing this imbalance may be important in the modeling phase to ensure that predictive models are not biased toward the majority class.

In the analysis of the “location.x” variable, we begin by examining its histogram and boxplot. The boxplot and histogram of the “location.x” variable provide insights into where shots are typically taken from in terms of field length. The histogram (Figure 2) shows that most shots are concentrated in the attacking third of the pitch, particularly between the 90 and 115 units on the x-axis, which aligns positions close to the opponent’s goal. The distribution is right-skewed, indicating that very few shots are taken from deeper positions in the pitch.



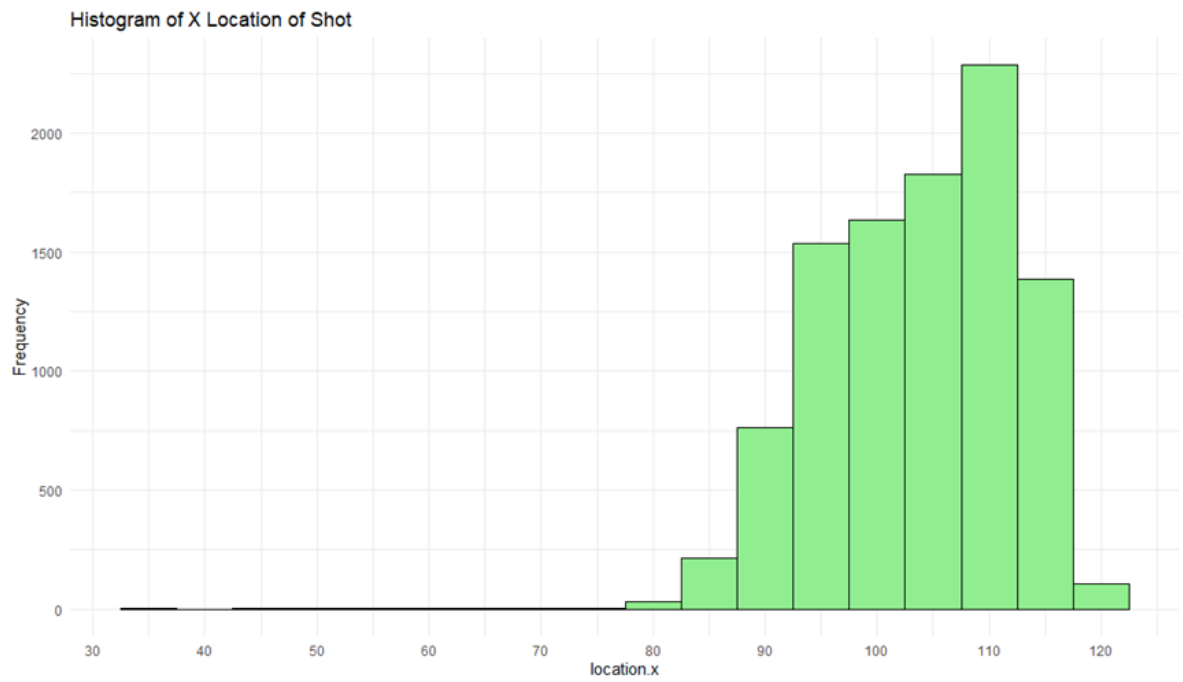


Figure 2: Histogram of “location.x” variable

The boxplot (Figure 3) complements this by illustrating the central tendency and spread of shot positions along the x-axis. It shows a high concentration of values near the upper quartile range, with a few noticeable outliers from deeper or unusual positions on the field. These outliers suggest rare long-range shot attempts or potentially irregular data points. Together, these plots confirm that most shots are taken from high x-coordinate areas, reinforcing the intuitive understanding that players predominantly shoot when close to the opponent's goal

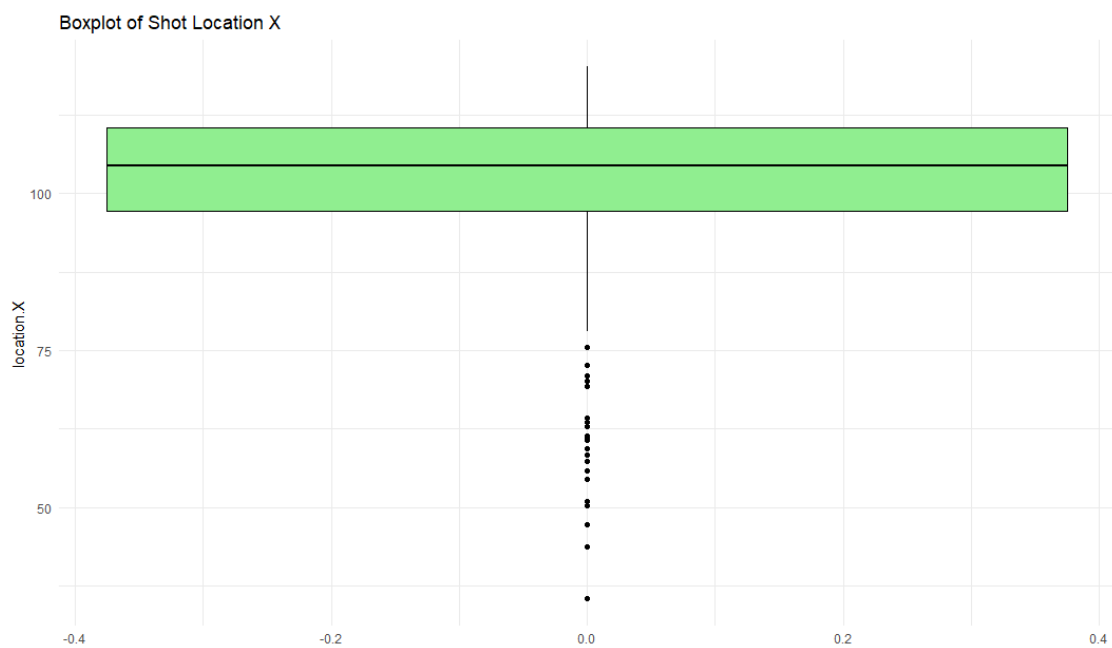


Figure 3: Boxplot of “location.x” variable



Following the examination of “location.x”, the distribution of “location.y” which represents the lateral (width) position on the pitch offers insights into shot placement across the field's width. The analysis of the “location.y” of shots provides insight into the horizontal distribution of shot attempts across the width of the pitch. The histogram (Figure 4) shows a clear concentration of shots near the central area, with the peak occurring around the midpoint of the field (approximately $Y = 40$), indicating that most shots are taken from central positions in front of goal. The distribution appears roughly symmetric, resembling a bell shape, which is expected in football, as players tend to shoot more frequently from central areas where the probability of scoring is higher, they see the goalposts more easily, and the central position of the shot is not require extreme technique always. The boxplot (Figure 5) further confirms this, showing a median near 40 and a relatively interquartile spread from about 30 to 50, with minimal outliers. These observations reinforce the strategic preference for central shooting positions during attacking plays.

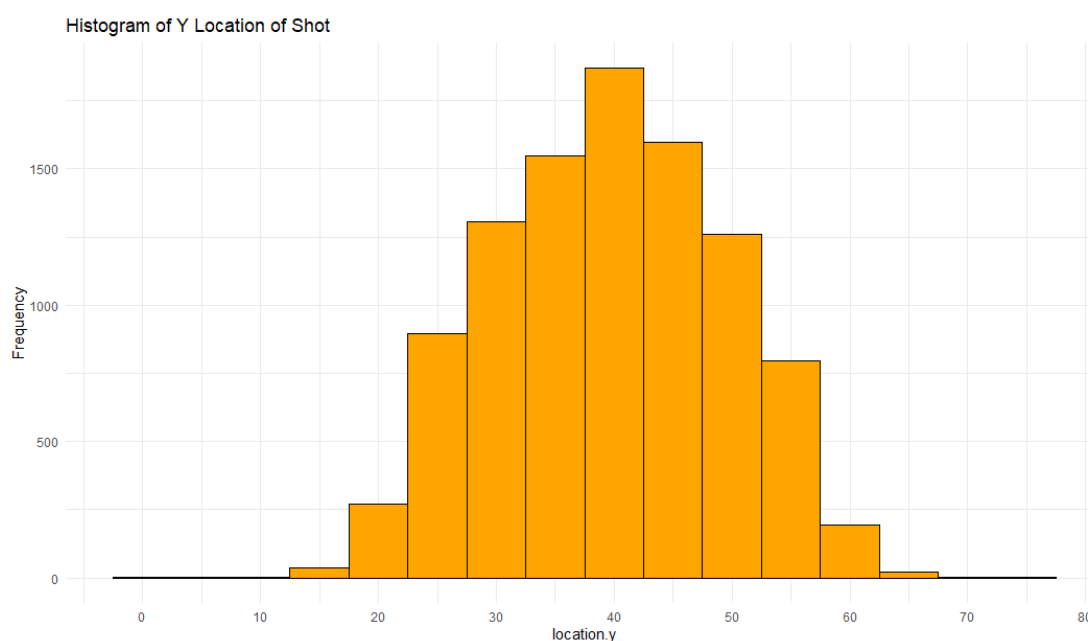


Figure 4: Histogram of “location.y” variable

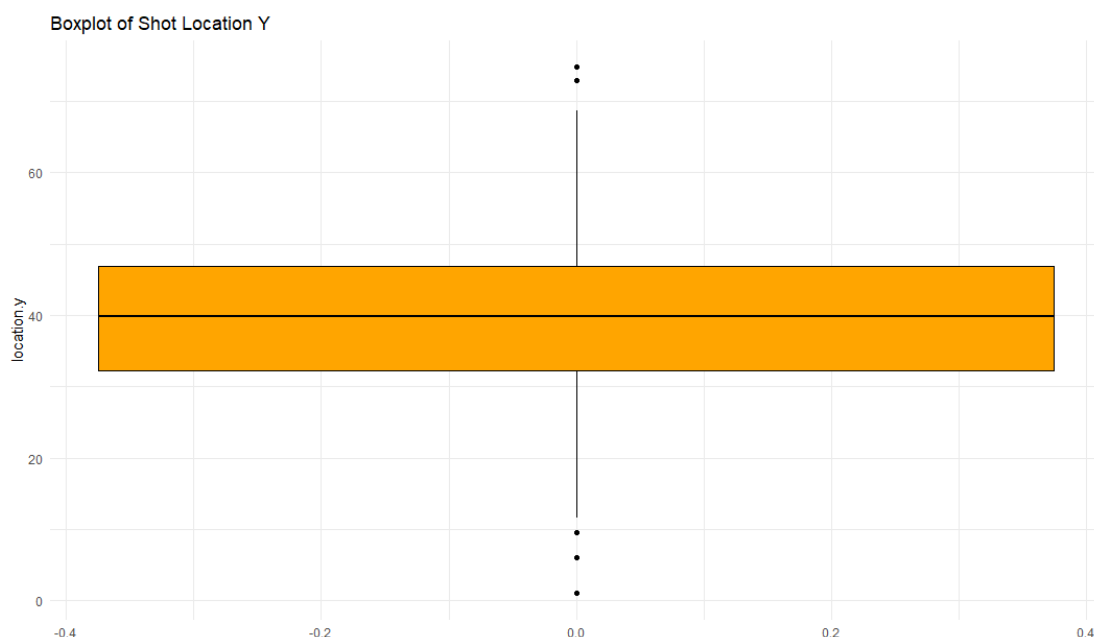


Figure 5: Boxplot of “location.y” variable

Following the univariate analysis of the “location.x” and “location.y” variables through histograms and boxplots, we now explore spatial patterns using shot density maps. These heatmaps offer a more intuitive view of where shots and goals are most frequently taken on the pitch. In the first heatmap (Figure 6), which includes all shot attempts, we observe a high concentration of activity around the central attacking zone, particularly between the penalty area and the goal line—an expected result given this is the most threatening area.

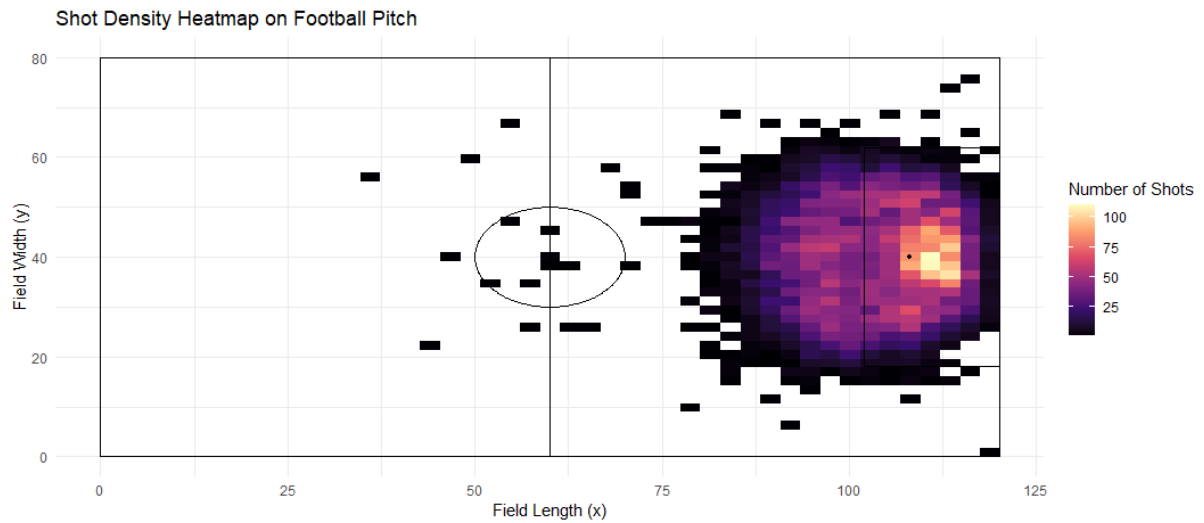


Figure 6: Shot Density Heatmap on Football Pitch- Exact locations of shots

The second heatmap (Figure 7), limited to shots that resulted in goals, reveals an even narrower hotspot. Most goals are concentrated directly in front of the goal and slightly toward the center, indicating higher scoring efficiency from this zone. This spatial insight reinforces the importance of central positioning and proximity to the goal in successful shot attempts and can inform tactical decisions for both offensive and defensive strategies.

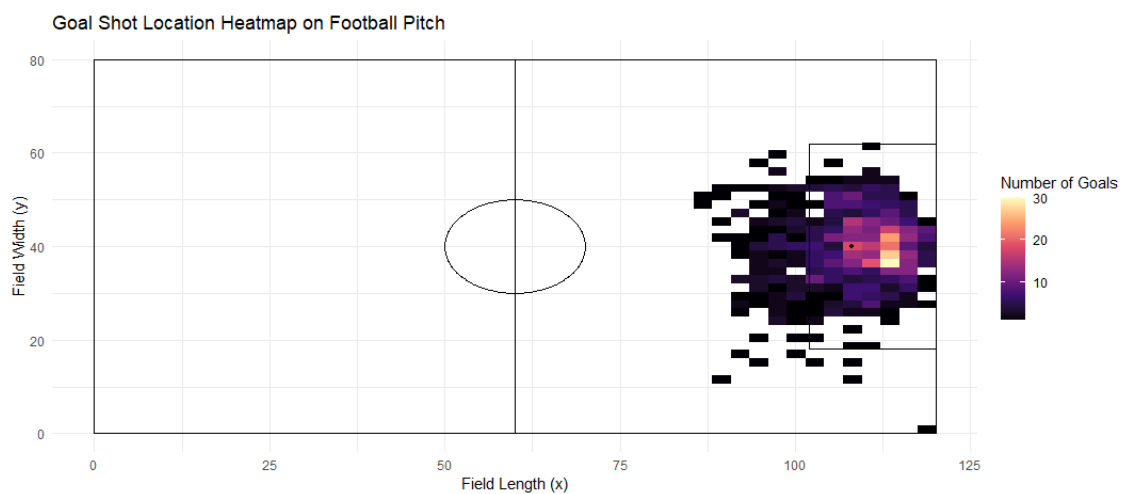


Figure 7: Goals Density Heatmap on Football Pitch- Exact locations of Goals

To continue the analysis of shot locations, we now turn our attention to the average velocity of the shots, an important quantitative variable that can influence the outcome of a goal attempt. To better understand the distribution and range of shot speeds, we utilize both a histogram and a boxplot. These visualizations allow us to observe the general spread of the data, identify the most frequent velocity intervals, and detect the presence of any extreme values or outliers. The histogram of shot average velocity (Figure 8) reveals a right-skewed distribution, with the majority of shots concentrated between 10 and 35 units of velocity. This suggests that most shots are taken at moderate speeds. However, there is a noticeable long tail extending toward higher velocity values, indicating that a smaller number of shots are taken at significantly higher speeds. The boxplot (Figure 9) further confirms this observation, showing a compact interquartile range and a large number of outliers above the upper whisker. These outliers represent exceptionally fast shots, which are uncommon but still present in the dataset. The combination of both plots highlights that while most shots fall within a typical speed range, a non-negligible number of high-velocity attempts may influence the dynamics of goal-scoring opportunities.

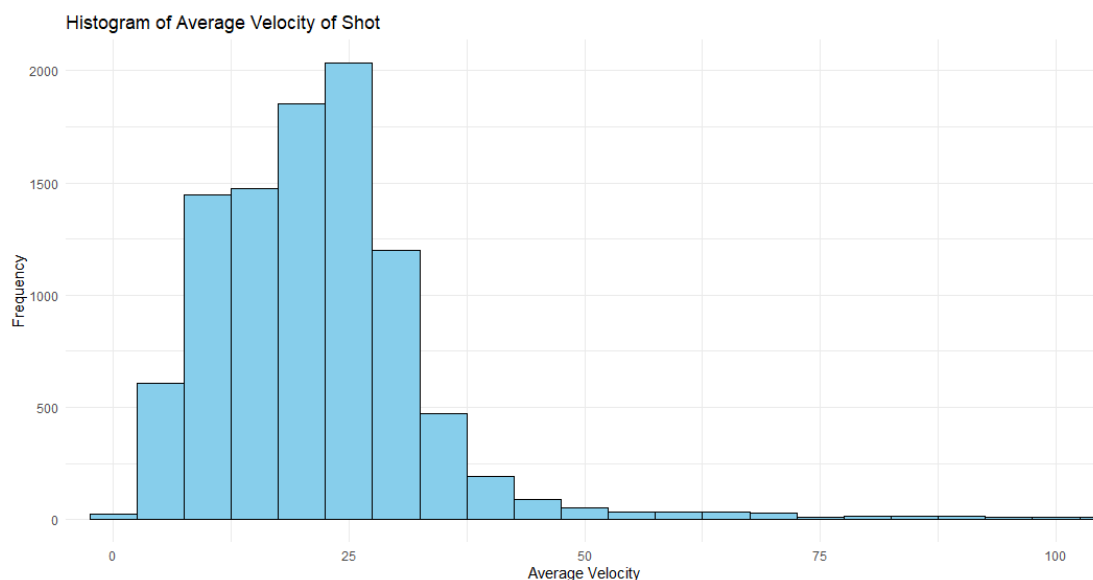


Figure 8: Histogram of Average Velocity variable

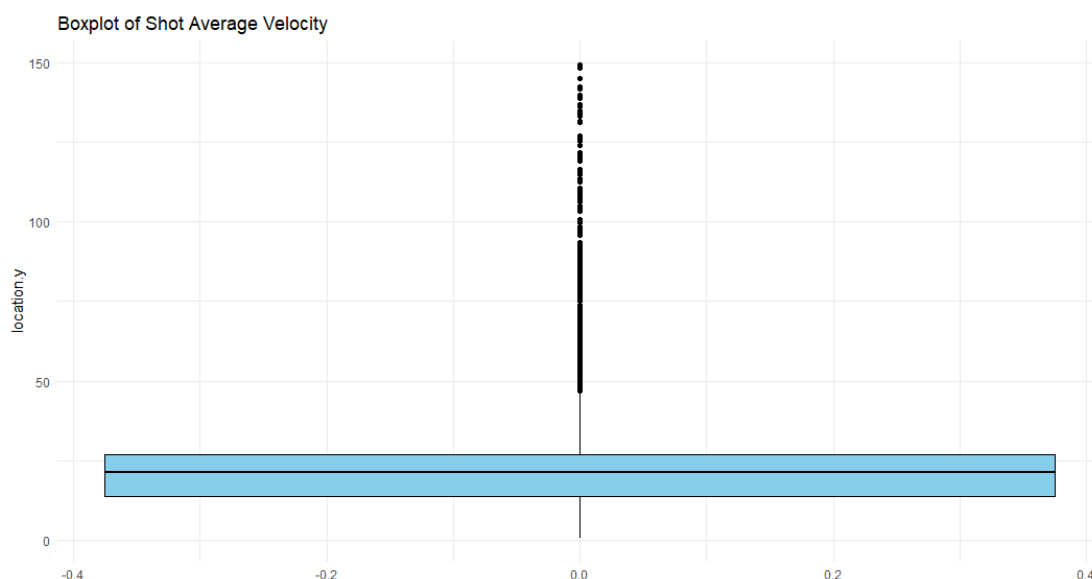


Figure 9: Boxplot of Average Velocity variable

After examining some of numerical variables, the next step in the univariate analysis focuses on categorical (factor) variables. These variables provide important qualitative insights about the nature of each shot, such as the technique used, the body part involved, the shot type, and whether the player was under pressure. By analyzing the frequency distribution of these factors, we can identify common patterns, preferences in playing styles, and contextual influences that may affect shot outcomes. Bar plots and frequency tables will be used to visualize and summarize the distribution of these variables, enhancing our understanding of shot characteristics across the dataset.

The barplot below (Figure 10) displays the distribution of the “under_pressure” variable, indicating whether a player was under defensive pressure at the moment of taking a shot. From the visualization, it is evident that the majority of shots (7,670 instances) were taken without pressure, while only 2,134 shots were taken under pressure. This disparity suggests that most shooting opportunities arise in less contested situations, which may impact shot quality and accuracy. The variable appears to be imbalanced, something that should be considered during modeling or feature evaluation.

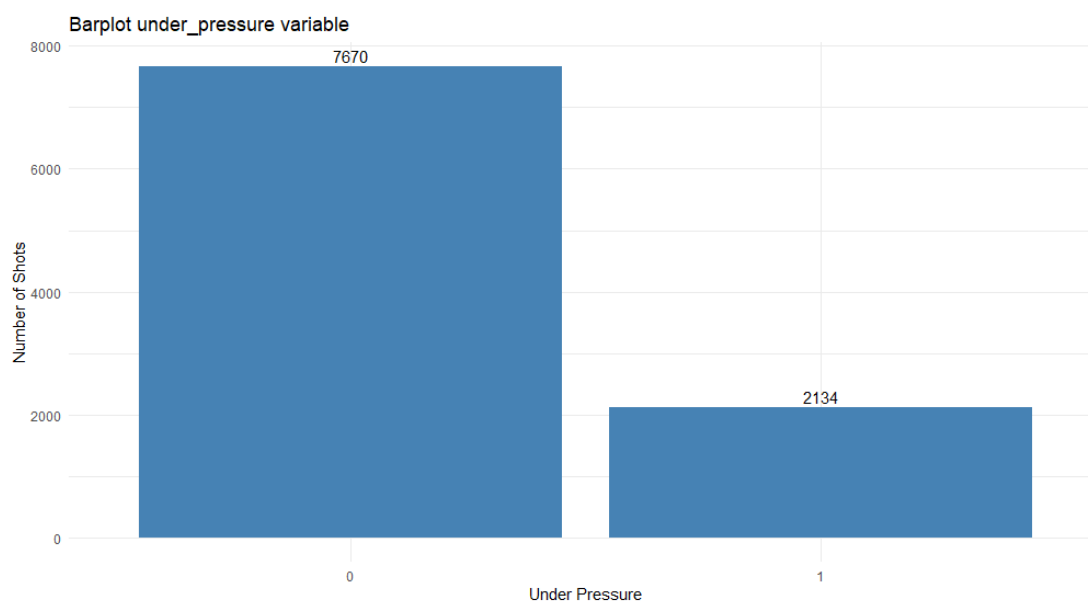


Figure 10: Barplot of “under_pressure” variable

Building on the categorical analysis, it is also valuable to investigate the role of player positioning in shooting behavior. The barplot in Figure 11 illustrates the distribution of shots by player position. Unsurprisingly, Center Forwards (Strikers and Second Strikers) lead in shot frequency, followed by Left Wings and Center Attacking Midfielders positions typically tasked with offensive duties. These roles naturally have more opportunities to shoot due to their advanced placement on the pitch and involvement in attacking plays. On the other hand, defensive and support roles such as Fullbacks, Defensive Midfielders, and Goalkeepers contribute far fewer shots, aligning with their tactical responsibilities. Understanding these positional patterns is essential, as position may serve as a strong predictor of shot volume and possibly shot success in later modeling phases.

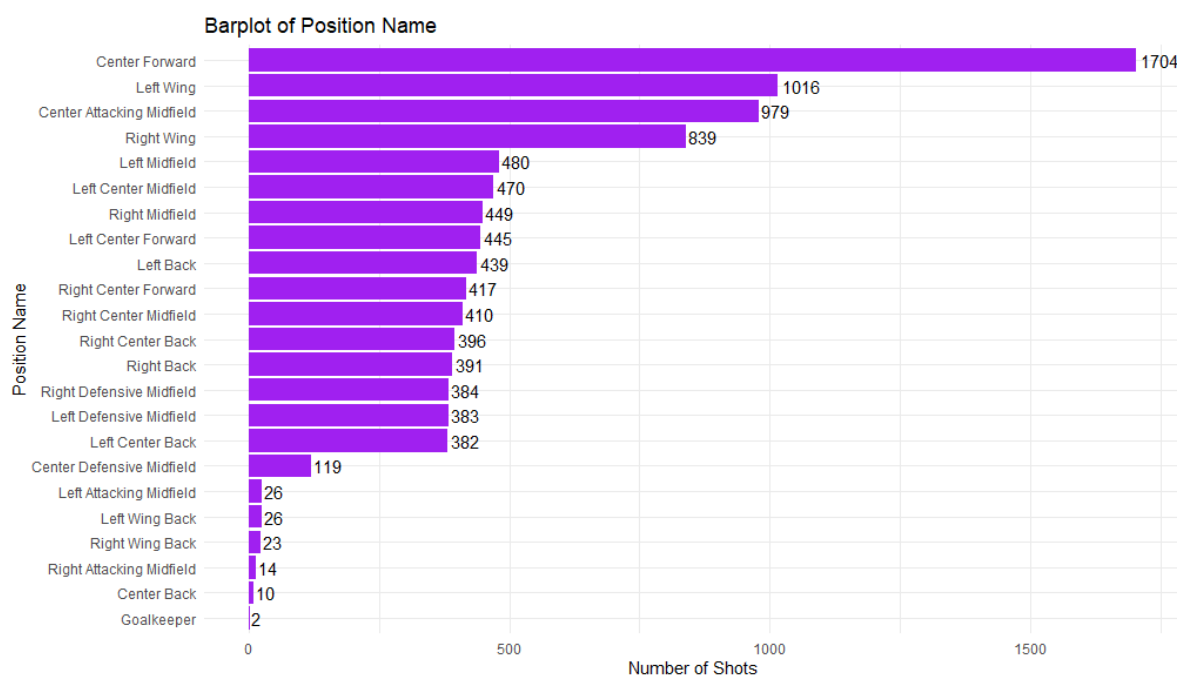


Figure 11: Horizontal barplot of positions

Further exploring categorical characteristics, the body part used to execute the shot provides key insight into player tendencies and technical preferences. As shown in the barplot in Figure 12, the right foot is by far the most frequently used body part, accounting for 5,285 shots, followed by the left foot with 2,895 shots. Shots taken with the head are considerably fewer (1,607), while the "Other" category is almost negligible (17 observations), likely capturing rare or misclassified events. This distribution aligns with the general prevalence of right-footed players in football and the dominance of foot-based finishing over headers. The clear imbalance across categories may also play a role in model performance if body part is included as a feature, highlighting the need for thoughtful treatment of rare levels during preprocessing.

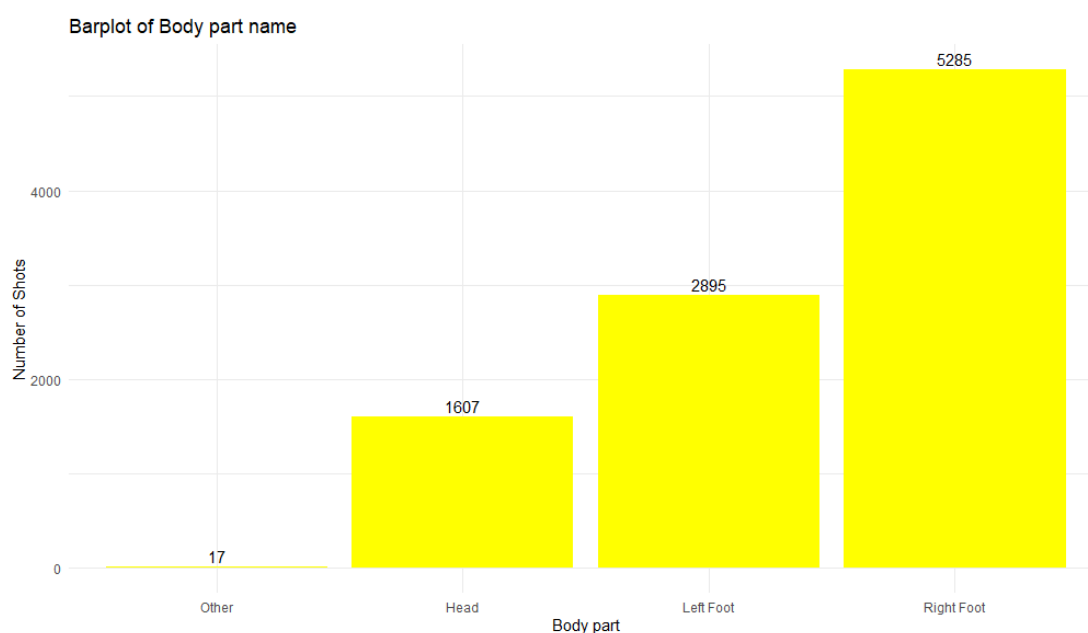


Figure 12: Barplot of Body part that shots made

4.4 Bivariate Analysis of Variables

Following the univariate exploration, the next step in the analysis involves examining bivariate analysis, relationship between variables. Bivariate analysis aims to explore how two variables relate to or influence each other, helping to uncover patterns, correlations, or potential predictors of the outcome variable — in this case, whether a shot results in a goal. By analyzing numerical and categorical variables in pairs, we can evaluate how specific features such as shot distance, body part used, or pressure status may impact shot success. This step is crucial for identifying meaningful associations and guiding feature selection for predictive modeling. The techniques employed include grouped bar plots, boxplots, correlation analysis, and cross-tabulations depending on the type of variables being compared (numerical vs. categorical). To begin the numerical bivariate analysis, a correlation matrix was computed among selected continuous variables. To maintain clarity and avoid visual clutter, only correlations with an absolute value greater than 0.4 ($|r| > 0.4$) were retained and plotted. This threshold-based filtering ensures that only the most meaningful and impactful relationships between variables are highlighted, allowing for better pattern recognition without the noise of weaker associations.

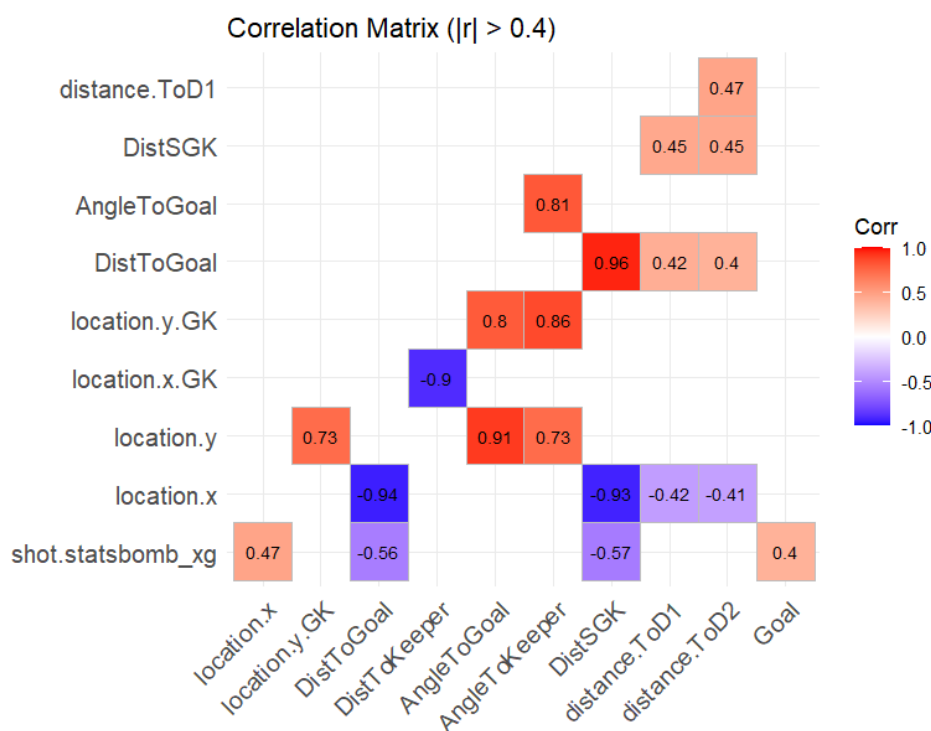
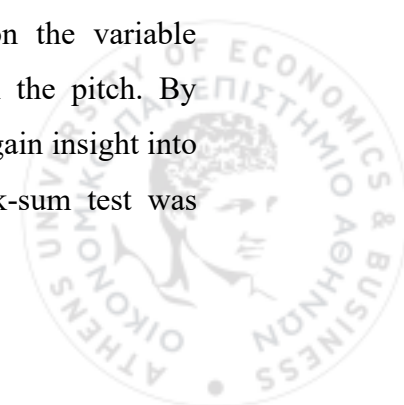


Figure 13: Correlation Plot of numeric variables that has correlation > 0.4

As shown in the filtered correlation plot (Figure 13), several strong relationships emerge. Notably, “AngleToGoal” is highly correlated with “AngleToKeeper” ($r = 0.81$) and this is logically right as in the most cases goalkeeper is in his “zone” so the angle to goal is almost the same with the angle from the goalkeeper. Similarly, the variables “DistToGoal” and “DistSGK” have presented high correlation ($r = 0.96$) for the same reason. On the other hand, we notice that we have also negative correlations. An instance of negative correlation is the correlation between variable “location.x” and “DistToGoal”. We can interpret this as, when the “location.x” is increasing, the Distance to goal is decreasing, as the player is more closely in the goal. These insights are crucial in guiding variable selection for predictive modeling, as they suggest that some predictors may convey overlapping information, while others offer independent predictive value. As a next step, care will be taken to address multicollinearity when fitting logistic regression models.

To further investigate relationships between key features and scoring outcomes, a bivariate analysis was conducted. The first comparison focuses on the variable “location.x”, which represents the horizontal position of a shot on the pitch. By examining how this variable differs between goals and non-goals, we gain insight into the role of shot proximity in scoring likelihood. A Wilcoxon rank-sum test was



conducted to evaluate whether the horizontal field position of a shot (location.x) differed between goals and non-goals (H_0 : the distribution of “location.x” is the same for shots that resulted in goals and those that did not). The test returned a highly significant result ($p < 2.2e-16$), indicating that null hypothesis is rejecting so there is difference in shot positioning between the two outcome groups. Specifically, shots that resulted in goals were taken from significantly lower x-coordinate values — that is, closer to the goal line. This finding aligns with football intuition and highlights the critical importance of shot proximity in scoring success. Figure 14 confirms this difference, as we can see that the goals are in a higher level of “location.x” which means that the player who took the shot is closer to the goal.

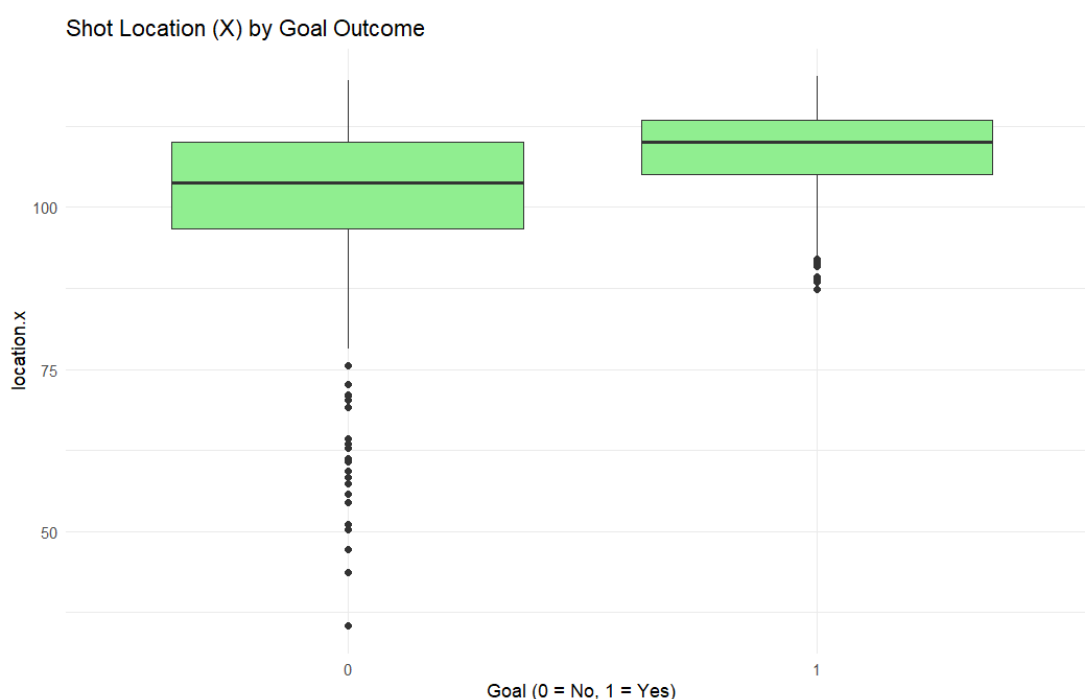
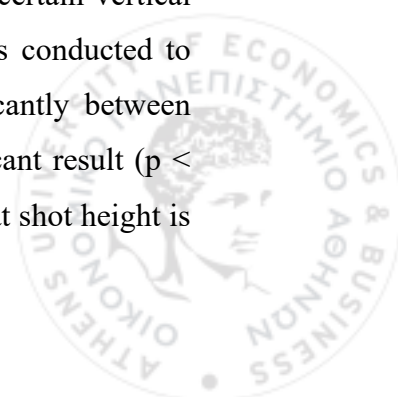


Figure 14: Boxplot of Goal and location.x variables

Continuing the bivariate analysis, the next variable explored is “shot.end_location.z”, which represents the vertical height (z-axis) at which the ball ended its trajectory — often interpreted as the height of the shot relative to the goal. Understanding how shot height varies between goals and non-goals helps to evaluate whether certain vertical zones are more effective for scoring. A Wilcoxon rank-sum test was conducted to determine if the distribution of “shot.end_location.z” differs significantly between successful and unsuccessful shots. The test produced a highly significant result ($p < 2.2e-16$), leading to a rejection of the null hypothesis. This suggests that shot height is



statistically different between goals and non-goals. As visualized in Figure 15, goals tend to have a lower shot height, with a noticeably lower median and tighter interquartile range. In contrast, non-goal shots show greater dispersion in height and a higher median value, including many high outliers. This pattern indicates that lower or mid-height shots are more likely to result in goals, potentially because they are more difficult for goalkeepers to reach or represent more controlled finishes. The result aligns with practical football understanding, where shots that are too high are more likely to be missed or be easier to be blocked.

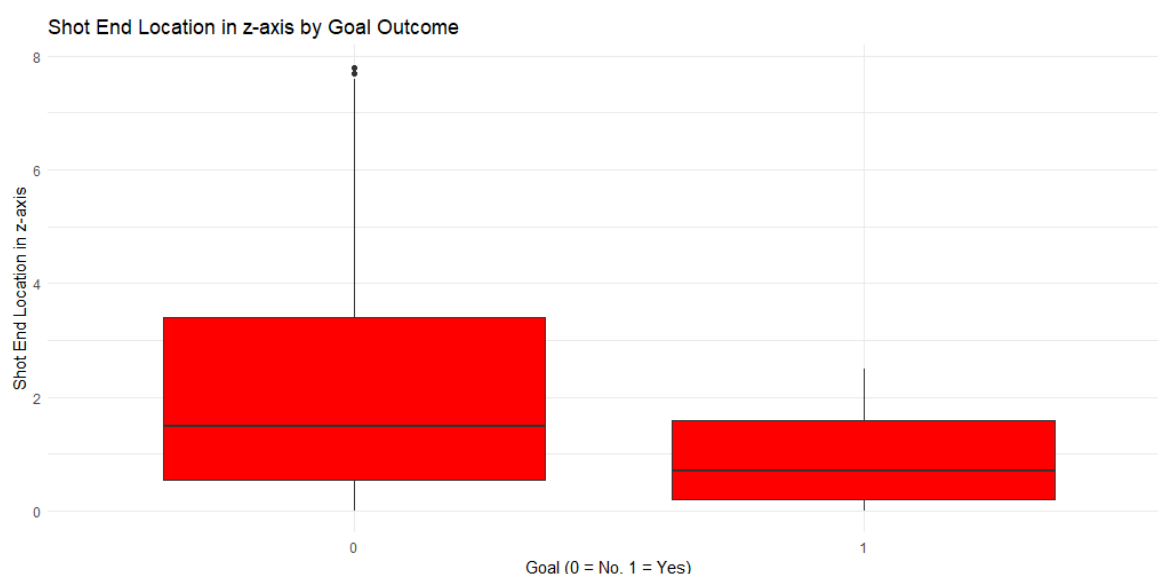


Figure 14: Boxplot of Goal and shot.end_location.z variables

In addition to continuous predictors, several categorical (factor) variables may influence the likelihood of a shot resulting in a goal. These include qualitative aspects such as the body part used to shoot, the player's position, or whether the player was under pressure. Exploring these variables provides insight into tactical patterns and player behaviors associated with successful finishing. For each factor variable, we assess the relationship with the binary outcome (Goal) using contingency tables, grouped bar plots, and appropriate statistical tests such as the Chi-squared test.

The variable “under_pressure” was analyzed to assess whether shooting under defensive pressure affected scoring outcomes. A Chi-squared test of independence was performed to evaluate the relationship between pressure and goal success. The result ($\chi^2 = 0.003$, $df = 1$, $p = 0.957$) indicates no significant association between the two variables. This suggests that, within this dataset, whether a player was under pressure

did not significantly impact their likelihood of scoring. Building upon the exploration of categorical variables, the next analysis focuses on the variable `technique.name`, which describes the method used to execute the shot—such as a volley, lob, or backheel. Understanding how different techniques influence goal outcomes can reveal tactical insights and player efficiency. Figure 15 presents a proportional stacked bar chart comparing the share of goals and non-goals across shot techniques. While the majority of shots across all techniques result in no goal, clear differences in scoring success emerge. Diving headers and lobbs exhibit notably higher goal proportions compared to techniques like normal shots, volleys, or overhead kicks. This suggests that certain techniques, though used less frequently, may be more effective in specific contexts, such as close-range or aerial situations.

These results indicate that shot technique has a meaningful influence on finishing success and should be considered in predictive modeling. Further statistical testing (e.g., Chi-squared test) could formally assess whether these differences are significant across groups. A Chi-squared test of independence was conducted to evaluate the relationship between `shot.technique.name` and scoring outcomes. The test returned a statistically significant result ($\chi^2 = 49.86$, $df = 6$, $p < 0.001$), suggesting that the technique used to execute a shot is significantly associated with whether it results in a goal. This implies that certain shot techniques are more effective than others in leading to goals.

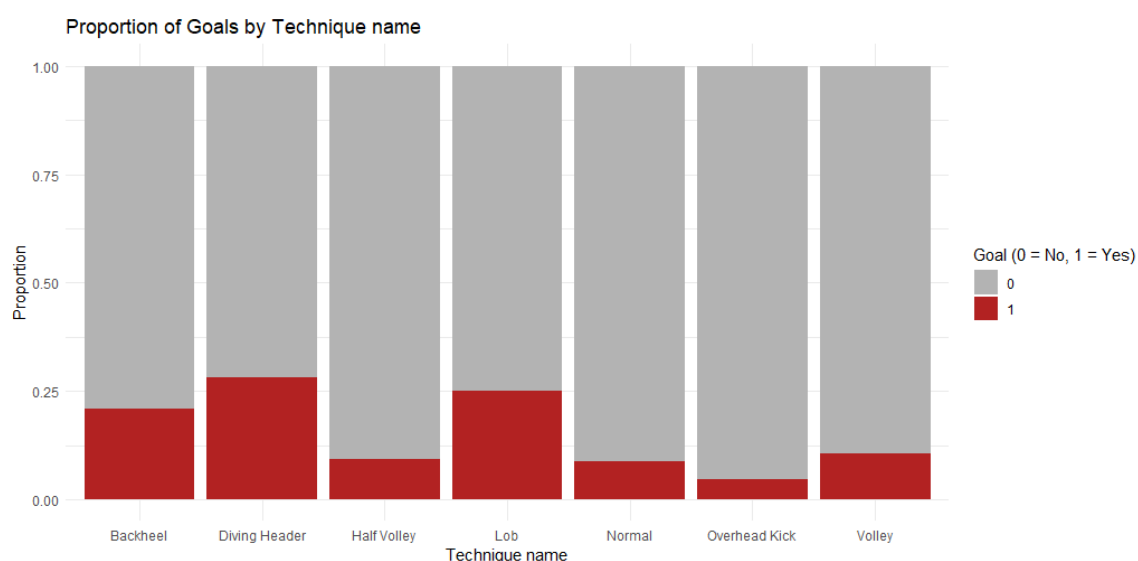


Figure 14: Stacked bar plot of Goal and technique name variables



The exploratory data analysis provided valuable insights into the characteristics and structure of the shot data, offering a solid foundation for predictive modeling. Univariate analysis revealed key trends in individual variables, such as the distribution of shot locations, velocities, and contextual factors like pressure and body part used. Bivariate analysis further deepened the understanding of relationships between features and the goal outcome. Notably, variables such as “location.x”, “shot.end_location.z”, and “DistToGoal” showed strong associations with scoring likelihood, confirmed through statistical testing and visualizations. Categorical factors, including shot.technique.name and body_part.name, also demonstrated significant variation in goal conversion rates. Together, these findings highlighted a set of meaningful predictors, reduced data noise, and informed the feature selection process for the xG modeling phase. Overall, the EDA underscored both the complexity and patterns within shot events, supporting the value of a data-driven approach to football analytics.



Chapter 5. Expected Goals Modeling

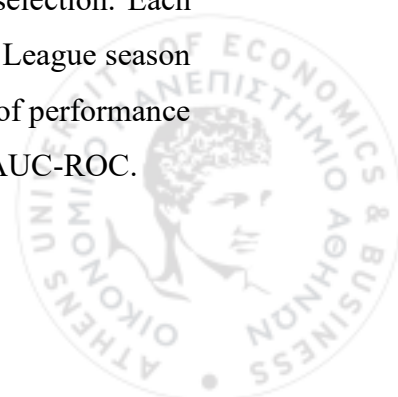
5.1 Model Selection: Logistic Regression

With a clear understanding of the data structure and feature relationships established through exploratory analysis, the final phase of this project involves developing statistical models to predict Expected Goals (xG). The objective is to build reliable models that estimate the probability of a shot resulting in a goal based on contextual and technical shot characteristics.

The main goal of this project is to predict the expected goal value, i.e. the probability of a shot to be a goal. As we want probability we thought about creating statistical models that will predict probability. The main statistical model selected for this prediction of goal outcomes in Expected Goals (xG) analysis is Logistic Regression. The main idea is the above thought. We have shots that have resulted in Goals or Non Goals. So as we have binary variable, Logistic regression is a classic tool for predicting binary outcomes (0 or 1), such as "Goal" (1) or "No Goal» (0). Unlike linear regression, which is suitable for predicting continuous variables, logistic regression uses the logit function to convert predictions into probabilities ranging from 0 to 1. More specifically, the model estimates the log odds of a shot resulting in a goal as a linear combination of independent variables, such as distance from the goal, angle of the shot, presence of defensive pressure, and goalkeeper position. The mathematical formula for logistic regression is given by the relationship:

$$\log\left[\frac{P}{1-P}\right]=\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k$$

The model uses p as the goal-scoring probability while X_i represents the independent variables. The expected goals application benefits from logistic regression through its ability to directly interpret coefficients as odds ratios. This chapter presents the modeling approach, starting with the full logistic regression model, followed by models refined through stepwise selection based on AIC and BIC criteria, as well as a LASSO-regularized model to address potential multicollinearity and variable selection. Each model is trained using the cleaned dataset from the 2015/2016 Premier League season and is evaluated against the actual goal outcomes using a consistent set of performance metrics, including accuracy, precision, recall, F1-score, deviance, and AUC-ROC.



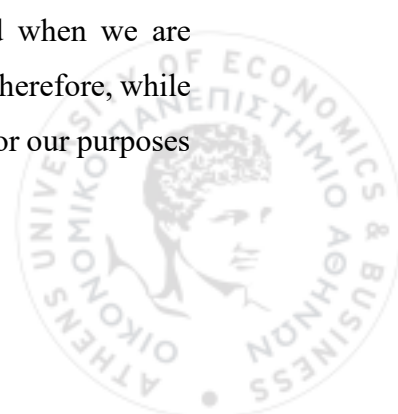
Additionally, model predictions are compared to the xG values provided by StatsBomb to benchmark the performance. This modeling process not only aims to produce accurate predictions but also to provide interpretable insights into which features most strongly influence scoring probability.

5.2 Model Training

Before training the predictive models, a final dataset was prepared by excluding specific variables that were either non-informative for prediction or could lead to data overfitting. These included “shot.statsbomb_xg” (an externally generated expected goals value), “type.name” and “shot.outcome.name” (which reveal the result of the shot directly), and “player.name” (to avoid overfitting to individual players). This resulted in a cleaned dataset (model_data) suitable for training.

To ensure objective evaluation of model performance, the dataset was randomly splitted into a training set (80%) and a testing set (20%) using a fixed seed (set.seed(123)) for reproducibility. The training set was used to fit the models, while the testing set was for evaluating predictive performance on data that the models has not seen during the training process.

In this project, four logistic regression models were developed to estimate the probability of a shot resulting in a goal (i.e., expected goals or xG). The process began with the construction of a full logistic regression model, including all available predictor variables after preprocessing. To improve the model and potentially improve generalizability, stepwise variable selection methods were then applied. Using the Akaike Information Criterion (AIC), a reduced model was generated by retaining only the variables that contributed to minimizing the AIC score. A similar procedure was followed using the Bayesian Information Criterion (BIC), which applies to a stricter penalty for model complexity, resulting in an even more parsimonious model. AIC and BIC while they may do the same job, differ in the scope of their use. AIC is usually used when we want to predict while in the other hand BIC is used when we are interested in finding the right factors that affect the model (Inference). Therefore, while both models are constructed, the BIC results may be more appropriate for our purposes in this case.



Finally, a fourth model was created using LASSO (Least Absolute Shrinkage and Selection Operator) regularization. This method performs both variable selection and coefficient shrinkage, making it especially useful in addressing multicollinearity and preventing overfitting. Each of these models was evaluated and compared based on their predictive performance and calibration using a consistent testing dataset. As expected, the full logistic regression model, which includes all available predictor variables, serves as a useful benchmark but lacks interpretability due to its complexity and the inclusion of potentially redundant or weakly informative features. While it may perform reasonably well in terms of predictive accuracy, its dense structure makes it less suitable for extracting meaningful insights.

Therefore, in the following analysis, greater emphasis is placed on the three refined models the AIC-selected, BIC-selected, and LASSO-regularized models. These models offer a more concise and interpretable representation of the key variables influencing scoring probability, making them more appropriate for tactical analysis and decision-making purposes.

5.3 Comparison and Evaluation of Logistic Regression Models

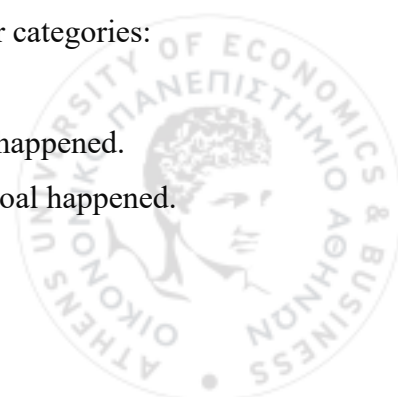
In order to improve predictive performance and assess model efficiency, four logistic regression models were evaluated using the final dataset:

- ***Model 1 – Full Logistic Model (all variables).***
- ***Model 2 – AIC-selected Model (stepwise AIC).***
- ***Model 3 – BIC-selected Model (stepwise BIC).***
- ***Model 4 – LASSO Logistic Model (L1-regularised).***

To effectively assess the performance of the predictive models developed in this project, several evaluation metrics were employed. Each metric captures a distinct aspect of the model's classification capability, helping to highlight strengths, weaknesses, and potential trade-offs. These metrics are all based on a tool called the Confusion Matrix, which breaks down the model's predictions into four categories:

True Positives (TP): The model correctly predicted a goal, and a goal happened.

True Negatives (TN): The model correctly predicted no goal, and no goal happened.



False Positives (FP): The model predicted a goal, but a goal did not actually happen.

False Negatives (FN): The model missed a goal, it predicted no goal, but a goal did happen.

Using these four values, we can calculate the main performance metrics.

Accuracy is the most used metric and measures the overall proportion of correct predictions both true positives (actual goals correctly predicted) and true negatives (non-goals correctly predicted).

While accuracy is useful when the dataset has a balanced class distribution, it can be misleading in cases where one class dominates such as when goals are much rarer than non-goals, so in our case accuracy may be misleading measure.

Precision refers to the proportion of predicted positive cases (goals) that were correct. This metric is especially important when the cost of false positives is high. In our context, a model with high precision means that when it predicts a goal, it is more likely to be correct, which is particularly useful in applications such as tactical analysis or betting systems.

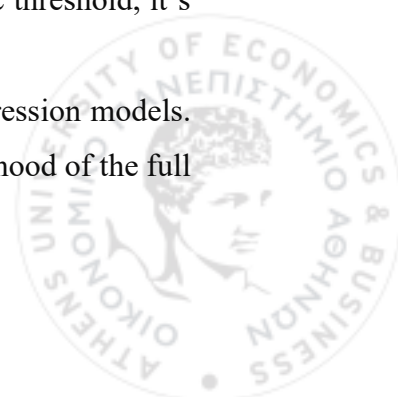
Recall, also known as sensitivity or the true positive rate, measures the ability of the model to correctly identify all actual goals.

A high recall score means the model is effective at identifying actual goals, even if it makes more false positive errors.

The F1 Score serves as a balance between precision and recall by computing their harmonic mean. It's especially helpful when the dataset is unbalanced, or when both false positives and false negatives are important. A high F1 Score means the model is doing a good job of correctly identifying real goals while also avoiding too many wrong predictions.

AUC, which stands for Area Under the ROC Curve, shows how well a model can tell the difference between positive and negative outcomes across all possible thresholds. The AUC score ranges from 0.5 (meaning the model has no real ability to separate the classes) to 1 (meaning perfect prediction). A model with an AUC higher than 0.9 is generally seen as excellent. Because AUC doesn't depend on a specific threshold, it's a useful way to compare how well different models perform overall.

Lastly, McFadden's R^2 is a pseudo- R^2 metric designed for logistic regression models. It assesses how well the model fits the data by comparing the log-likelihood of the full model to that of a null model (a model with only an intercept).



McFadden's R^2 values between 0.2 and 0.4 are generally seen as a sign of good model fit. While it's not directly comparable to the R^2 used in linear regression, it still gives a useful sense of how much better the model performs compared to a basic one with no predictors.

Taken together, these evaluation metrics provide a good way to judge how well a classification model works. Depending on the goal whether it's maximizing overall accuracy, reducing false positives, or catching as many true positives as possible some metrics may be more important than others. In this study, we used all of them to make sure our model assessment was thorough and well-rounded.

The table below summarizes the key evaluation metrics:

Model	Accuracy	Precision	Recall	F1 Score	AUC	McFadden R^2
Full Logistic	0.92	0.64	0.47	0.54	0.94	0.53
AIC-selected	0.92	0.65	0.45	0.53	0.94	0.5
BIC-selected	0.92	0.63	0.42	0.5	0.94	0.48
LASSO	0.92	0.67	0.48	0.56	0.94	0.53

Table 4: Model Evaluation Metrics

The evaluation metrics presented in Table 4 offer valuable insights into the predictive performance and trade-offs of each model. All four models demonstrate high and consistent accuracy, with values around 0.92, confirming their reliability in classifying goal outcomes. The Full Logistic model demonstrates reliable predictive performance, achieving a strong accuracy of 0.92 and a high AUC of 0.94 (Figure 15), indicating good overall classification capability. It delivers a precision of 0.64, a recall of 0.47, and an F1 Score of 0.54, placing it just behind the LASSO model in recall and F1 performance. Notably, the Full model achieves the highest McFadden R^2 (0.53) among all models, suggesting excellent explanatory power and model fit. However, this comes with reduced interpretability due to the large number of included variables, reinforcing the value of simpler alternatives like the AIC, BIC, or LASSO-selected models for practical use.



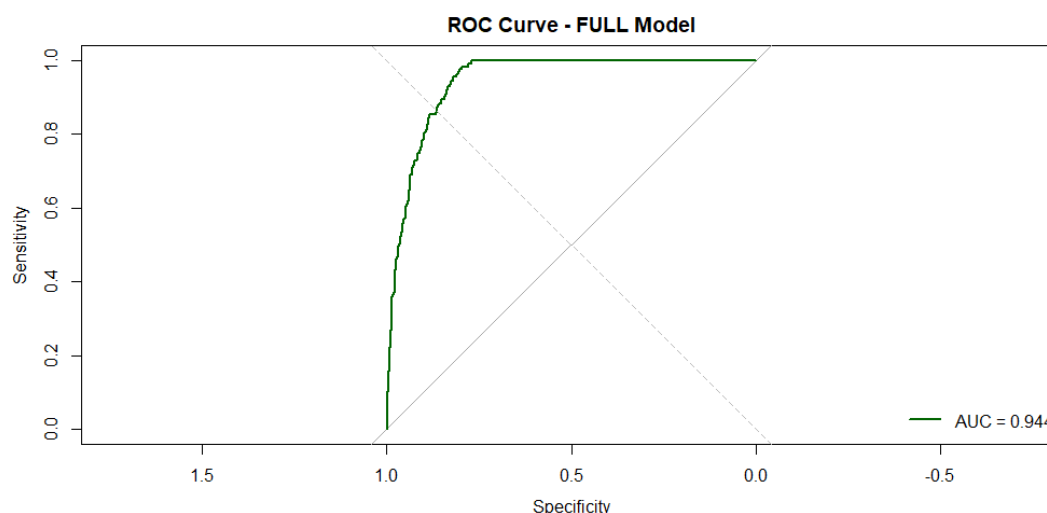


Figure 15: Receiver Operating Characteristic curve for Full model

The LASSO model shows the best precision (0.67) and F1 Score (0.56), indicating that its goal predictions are more often correct, though this comes with reduced recall (0.48). With high accuracy (0.92) and competitive AUC in 0.94 (Figure 16) , LASSO presents a strong case for use in scenarios where false positives must be minimized, such as in betting or conservative risk-based decisions.

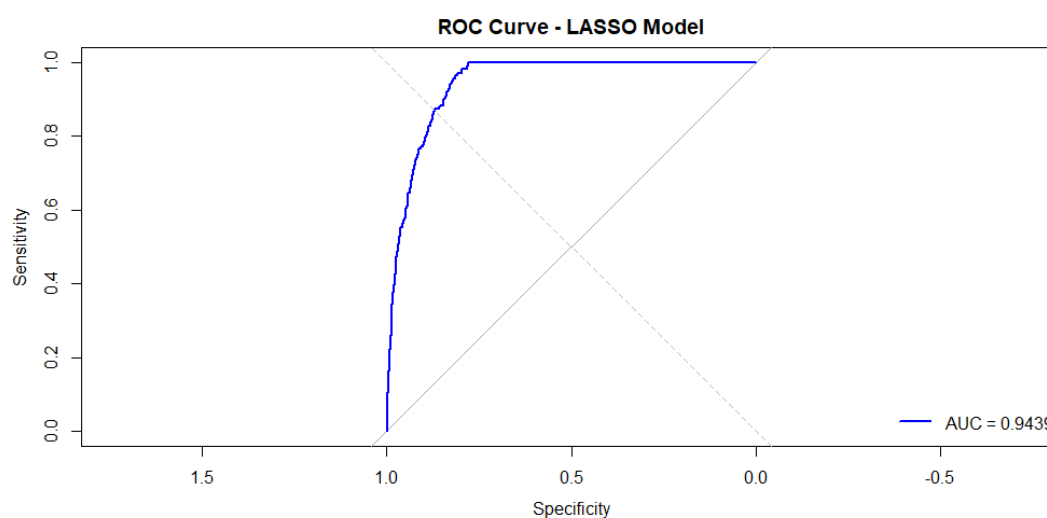
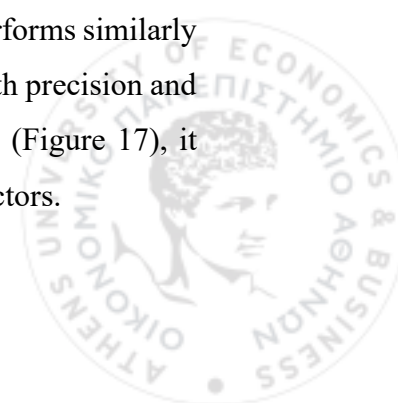


Figure 16: Receiver Operating Characteristic curve for Lasso model

The AIC-selected model remains well-balanced across all metrics. It performs similarly to the full model in accuracy and AUC, while improving slightly on both precision and model simplicity. With a McFadden R^2 of 0.50 and an AUC of 0.94 (Figure 17), it provides solid explanatory power with a more manageable set of predictors.



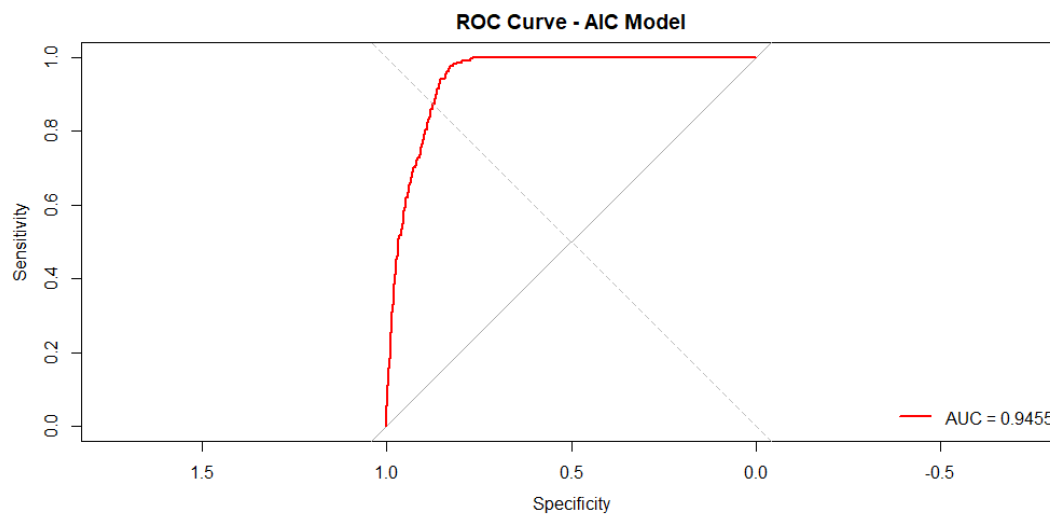


Figure 17: Receiver Operating Characteristic curve for AIC model

The BIC-selected model, although slightly lower in recall (0.42), retains a respectable F1 Score (0.5), high precision (0.63), and a strong AUC 0.94, making it an interpretable and compact alternative. Its reduced complexity enhances generalizability while maintaining predictive strength.

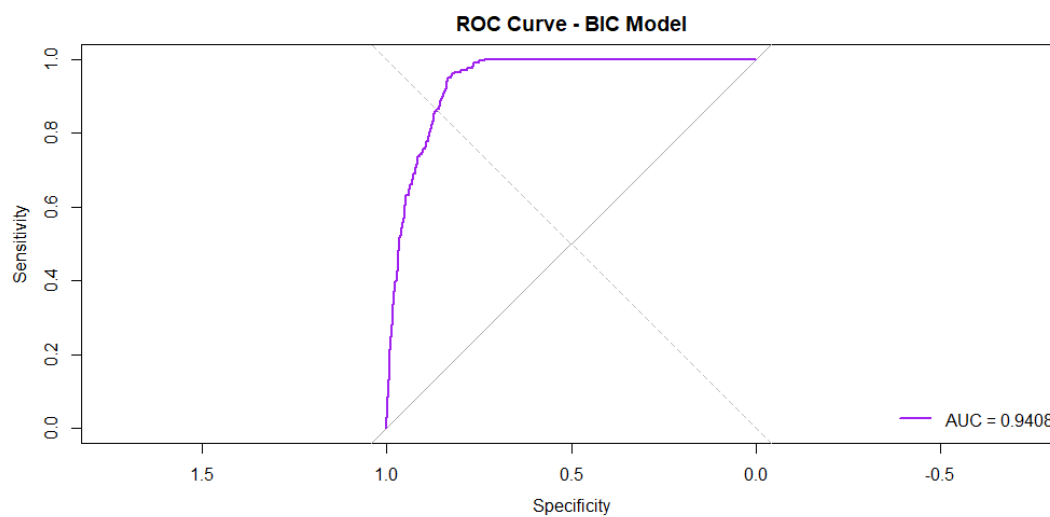
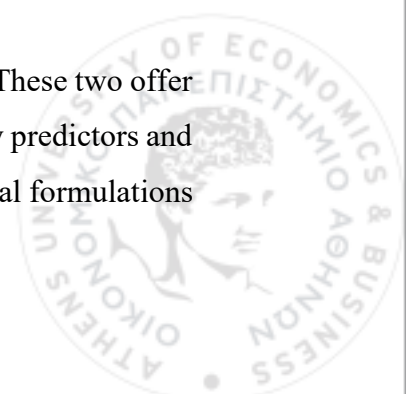


Figure 17: Receiver Operating Characteristic curve for BIC model

Given the trade-offs, we narrow our focus to the AIC and BIC models. These two offer the best balance between performance and interpretability, retaining key predictors and discarding noise. In the sections that follow, we present the mathematical formulations



of both reduced models and examine the influence of statistically significant variables on goal probability.

Mathematical Equations of the Models

$$\begin{aligned} \text{logit}(P(\text{Goal}=1)) = & -836.93 - 0.28 \cdot \text{under_pressure1} + 1.80 \cdot \text{shot.technique.nameDiving Header} + 1.29 \cdot \text{shot.body_part.nameLeft Foot} \\ & + 2.40 \cdot \text{shot.body_part.nameOther} + 1.30 \cdot \text{shot.body_part.nameRight Foot} - 0.11 \cdot \text{location.x} - 0.03 \cdot \text{location.y} \\ & + 7.41 \cdot \text{shot.end_location.x} - 0.70 \cdot \text{shot.end_location.z} - 0.13 \cdot \text{location.x.GK} - 0.27 \cdot \text{DistToGoal} - 0.23 \cdot \text{InCone.GK} \end{aligned}$$

Formula 1: Model after AIC-selection

The above equation represents the logistic regression model selected using the AIC criterion, where the log-odds of a shot resulting in a goal ($\text{logit}(P(\text{Goal}=1))$) is modeled as a linear function of several predictor variables. Each coefficient reflects the magnitude and direction of influence that the corresponding variable has on the likelihood of scoring.

- The intercept (-836.93) express the baseline log-odds when all predictors are zero.
- The variable `under_pressure1` has a negative coefficient (-0.28), suggesting that shots taken under pressure are associated with a lower probability of scoring.
- Techniques like "Diving Header" (+1.80) increase the log-odds of scoring, implying they are more likely to result in goals than the reference category.
- Shot body parts also have notable influence: for example, using the left foot (+1.29) or right foot (+1.30) both increase scoring likelihood relative to the baseline, while "Other" body parts (+2.40) have the strongest positive association.
- Positioning variables such as `location.x` (-0.11), `location.y` (-0.03), and `shot.end_location.z` (-0.13) suggest that as the shot is taken from wider or higher vertical angles, the likelihood of a goal decreases.
- A closer shot ending horizontally (`shot.end_location.x` = +7.41) significantly increases goal probability — aligning with football intuition.



- Additionally, goalkeeper-relative variables such as location.x.GK (-0.70) and InCone.GK (-0.23) negatively influence the odds, indicating that certain shot angles or goalkeeper positioning reduce scoring chances.
- Lastly, DistToGoal (-0.27) confirms that greater shot distance from the goal lowers the probability of a successful outcome.

Altogether, this model captures both player characteristics and positional dynamics, offering a statistically grounded and interpretable framework for understanding goal likelihood. It underscores the importance of proximity, technique, and pressure in determining scoring success.

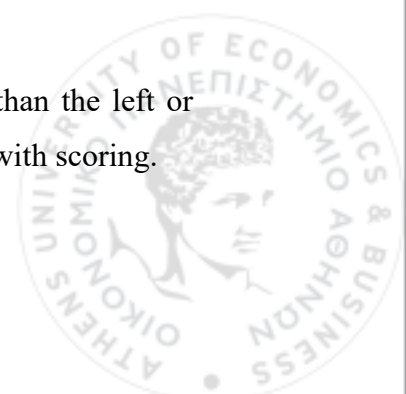
Following the analysis of the AIC-selected model, we now turn our attention to the logistic regression model selected using the Bayesian Information Criterion (BIC). The BIC model is constructed with a stricter penalty on model complexity compared to AIC, favoring more brief models. As a result, it retains only the most essential predictors that contribute meaningfully to the prediction of goal outcomes.

$$\begin{aligned} \text{logit}(P(\text{Goal}=1)) = & -829.24 + 0.98 \cdot \text{shot.body_part.nameLeft Foot} + 2.50 \cdot \text{shot.body_part.nameOther} \\ & + 1.03 \cdot \text{shot.body_part.nameRight Foot} - 0.06 \cdot \text{location.x} + 7.14 \cdot \text{shot.end_location.x} \\ & - 0.67 \cdot \text{shot.end_location.z} - 0.16 \cdot \text{location.x.GK} - 0.19 \cdot \text{DistToGoal} \end{aligned}$$

Formula 2: Model after BIC-selection

The BIC model retains only eight variables, highlighting the most critical factors associated with scoring outcomes.

- **Intercept (-829.24):** The baseline log-odds of scoring when all predictor variables are zero. As expected in logistic models with continuous predictors, the intercept is not directly interpretable alone.
- **shot.body_part.nameLeft Foot (0.98):** Shots taken with the left foot increase the probability of scoring.
- **shot.body_part.nameOther (2.50):** Using a body part other than the left or right foot (e.g., head or chest) has a strong positive association with scoring.



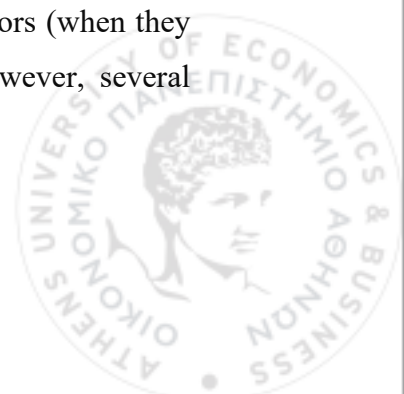
- **shot.body_part.nameRight Foot (1.03):** Shots with the right foot also increase the likelihood of a goal.
- **location.x (−0.06):** Shots taken from positions farther up the pitch (toward midfield) slightly reduce scoring chances.
- **shot.end_location.x (7.14):** Shots that end closer to the goal line (higher x-coordinate) are significantly more likely to result in goals.
- **shot.end_location.z (−0.67):** Higher shot placements (on the vertical axis) are associated with a lower probability of success.
- **location.x.GK (−0.16):** As the goalkeeper's x-position moves further from the shooter, the chance of scoring slightly decreases.
- **DistToGoal (−0.19):** Greater overall distance from the goal reduces the scoring probability.

The final model used in this project was created through the LASSO (Least Absolute Shrinkage and Selection Operator) procedure. This method helps to reduce model complexity by shrinking the coefficients of less important variables to zero, effectively removing them from the model. As a result, the LASSO model includes only the most important predictors and avoids overfitting. This makes it easier to interpret and more reliable for general use. In the next section, we present the mathematical formula of the LASSO-selected model and explain the role of each variable included.

$$\begin{aligned} \text{logit}(P(\text{Goal}=1)) = & -862.21 - 0.28 \cdot \text{under_pressure1} + 1.19 \cdot \text{shot.body_part.nameLeft Foot} + 2.49 \cdot \text{shot.body_part.nameOther} \\ & + 1.17 \cdot \text{shot.body_part.nameRight Foot} - 0.09 \cdot \text{location.x} + 1.80 \cdot \text{shot.technique.nameDiving Header} \\ & + 7.67 \cdot \text{shot.end_location.x} - 0.74 \cdot \text{shot.end_location.z} - 0.03 \cdot \text{location.y} \\ & - 0.17 \cdot \text{location.x.GK} - 0.26 \cdot \text{DistToGoal} \end{aligned}$$

Formula 3: Model after LASSO

The LASSO-selected logistic regression model includes a focused set of predictors, each contributing to the likelihood of a goal being scored. The intercept is very large and negative (−862.21), which means that in the absence of all predictors (when they are zero), the baseline probability of scoring is extremely low. However, several variables increase or decrease this probability depending on their value.



- Being under pressure during the shot slightly lowers the chance of scoring (coefficient: -0.28), likely due to reduced control or accuracy.
- Shots taken with the left foot ($+1.19$), right foot ($+1.17$), or with other body parts ($+2.49$) significantly increase the scoring probability compared to the reference category
- The Diving Header technique also has a positive effect ($+1.80$), possibly due to its use in closer or well-placed attacking situations.
- Shot end location x ($+7.67$) indicates that shots ending closer to the goal (higher x value) are much more likely to result in a goal.
- Shot end location z (-0.74) shows that shots finishing higher in the air are less likely to score, which aligns with shots going over the crossbar.
- The horizontal (x) and vertical (y) locations of the shot's origin are also included: location. x ($+0.09$) slightly increases goal chances, while location. y (-0.03) has a minimal negative effect.
- Goalkeeper's x -position (location. x .GK: -0.17) and distance to goal (DistToGoal: -0.26) both negatively affect the probability, which makes sense—if the goalkeeper is well-positioned or the shot is taken from further away, the likelihood of scoring drops.



Chapter 6. Conclusions and Discussion

6.1 Key findings and their significance

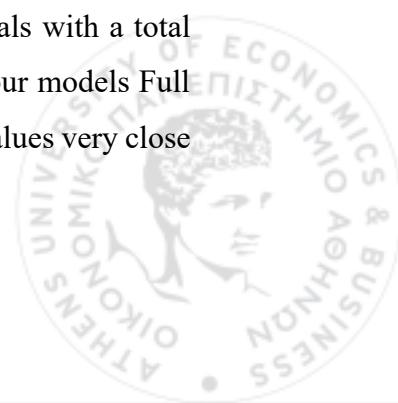
By using data from the 2015/2016 Premier League competition, we studied many important factors that may affect shot outcomes, such as, distance from goal, angle from goal, type of final effort (head, left foot, right foot, etc.). After a detailed data cleaning and exploratory data analysis that conducted with plots and tables, 4 logistic regression models created including Full model, model after AIC and BIC criteria and a LASSO model.

The main idea and scope of this analysis was to create statistic models that will predict effectively the xG indicator of a shot. In our dataset as we have referred above in the report, we had a column named “shot.statsbomb_xg” which contains the xG indicator value from the data provider (StatsBomb). Initially, when we started to train the models we had excluded this column from the dataset in order to create this indicator and the models to be unaffected from this column. So, when we returned in the test dataset we thought, after the creation of models and apart from the evaluation metrics (Accuracy score, AUC etc.) to compare our results with this column in order to see how well our models work. This comparison confirmed that our statistical models give satisfying predictions.

Model	Total Goals / xG
Actual Goals	197.0
xG StatsBomb	186.49
xG Full Model	197.14
xG AIC Model	197.22
xG BIC Model	196.6
xG LASSO Model	197.19

Table 5: Results- Estimations of models in test dataset

The table above compares the total number of goals in the test dataset with the expected goals (xG) predicted by different models. The actual number of goals recorded in the test dataset was 197. The StatsBomb xG model underestimates the goals with a total xG of 186.49, suggesting a more cautious prediction. In contrast, all four models Full Model, AIC Model, BIC Model, and LASSO Model produce total xG values very close to the actual number of goals, with deviations of less than 0.40.



Among the models tested, the AIC and LASSO models came closest to predicting the actual number of goals (197.22 and 197.19), which shows they worked well and gave reliable results. The BIC Model, while underestimating at 196.60, still performs very well. This close alignment indicates that the custom statistical models, particularly those selected through regularization and information criteria, provide more accurate aggregate predictions compared to the original StatsBomb xG. This supports the usefulness of tailored model development in football analytics.

Moreover, for the factors that models provided now. We see in our models that some factors, as it is logical, are more affecting in the shot outcome. We notice in all models as it was expected that the location of the player that takes the shot and the location of the Goalkeeper are two very vital variables. It is extremely interesting also that the body part of final effort is an important factor. We inferred above in paragraph 2.2.2 that the type of final effort plays a crucial role in the outcome. It will be very interesting if we have also the information about strong/weak foot. For instance, to have the information of each player if he takes the shot with the strong foot (binary variable). Finally, LASSO model contains both of “location.x” and “location.y” which expresses the exact location of the player that takes the shot. With this combination, in the model included also the information of angle from the goal, not only the distance.

Overall, this project supports the idea that data analysis and xG models are valuable tools in modern football. They can help coaches, analysts, and even betting companies by offering clearer and more accurate insights into team and player performance. Still, it's important to remember that limitations like missing data or the unpredictable nature of the game can affect the accuracy of results. Also, we must not forget that football is based on rare events. Looking ahead, using more advanced models and richer datasets could lead to even better predictions and help teams make smarter decisions based on data.



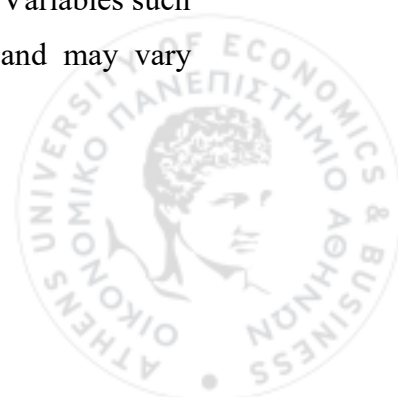
6.2 Suggestions for future research and limitations

As a first thought, a useful direction for future research is to test how well xG models work in different football leagues. So far, this study focused only on the English Premier League, but it would be interesting to see how these models perform with data from other top competitions like La Liga or Serie A. This would help show whether the same variables still work well for predicting goals or if the models need to be adjusted because of differences in playing styles, defensive tactics, or shot behavior. Learning how to adapt or retrain models for new data is important for making xG analysis more useful across different football settings. In this way we will be able to understand better the differences between different leagues and to notice if there are significant differences in xG indicators. Teams and coaching staffs will be helped from this as it will be easier to identify in which part-field of the game they underperform and will enforce their strategy more cleverly.

Additionally, future work could examine how expected goals metrics behave on an individual player level. It would be interesting to investigate whether xG values vary significantly from one player to another, even under similar conditions. Are certain players statistically more efficient in turning chances into goals? Is this difference significant across seasons or clubs? This kind of analysis can help show which parts of finishing ability stay consistent over time, giving us a better understanding of a player's quality and how reliable their performance is.

Lastly, it is very interesting to aim at the prior phases of shot. This project has analyzed the shots exactly at the moment that the shot was taken. If we had with some way the data prior to the shot we would be able to analyze also some factors like position of all players in the pitch or identify specific patterns of formations of teams.

One of the main limitations of this study is the nature of football itself. Football is a complex and dynamic sport, where many events are difficult to be predicted with high accuracy. While statistical models can capture certain patterns and trends, they cannot measure all factors that influence a match. For example, it is challenging to include the difference in quality between two teams in a simple and objective way. Variables such as team strength, tactics, or form are hard to measure accurately and may vary significantly from match to match.



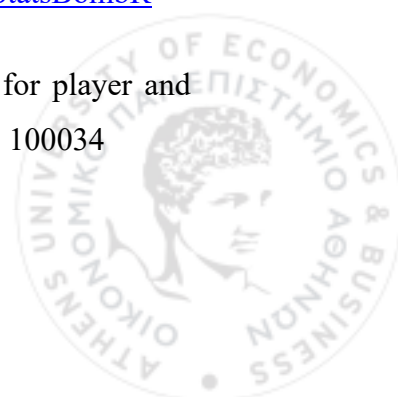
Another tricky part is the psychological side of football. Things like pressure, confidence, motivation, and even the energy from the fans can have a huge effect on how a game plays out. But those influences are difficult to turn into numbers and include in a dataset. Because of this, no matter how advanced our models get, there will always be a limit to how much they can explain or predict. Football remains a game full of surprises and that's part of what makes it so exciting.

End of Thesis Report



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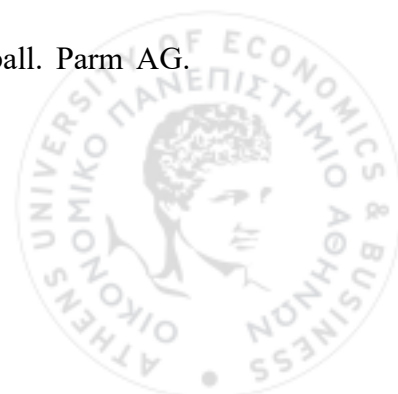
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Appendix: R Code for Statistical Analysis and Modeling

The appendix contains two R code files which were used for the practical implementation of the work “Modeling Expected Goals in Football”.

- **Data Extraction Script** – This script was used to extract the required data and save the specific dataset needed for the analysis.
- **Analysis Script** – This script contains all steps of the analysis, including data cleaning, modeling, and visualization.

